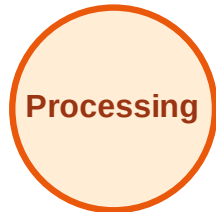


# BFS Traversal

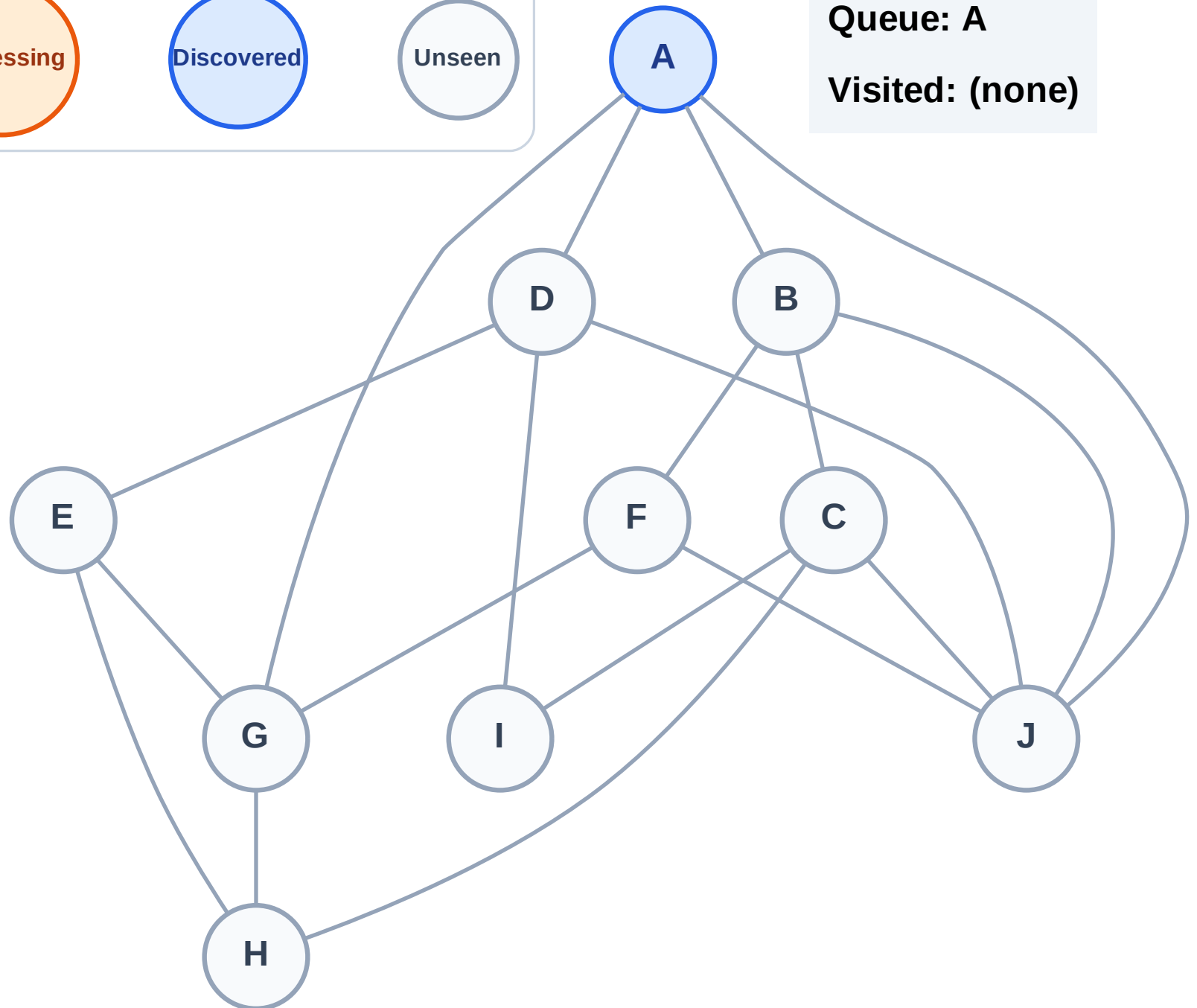
Step 1: Start at A

## Legend



Queue: A

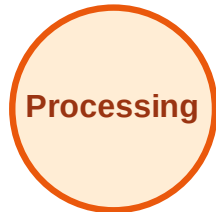
Visited: (none)



# BFS Traversal

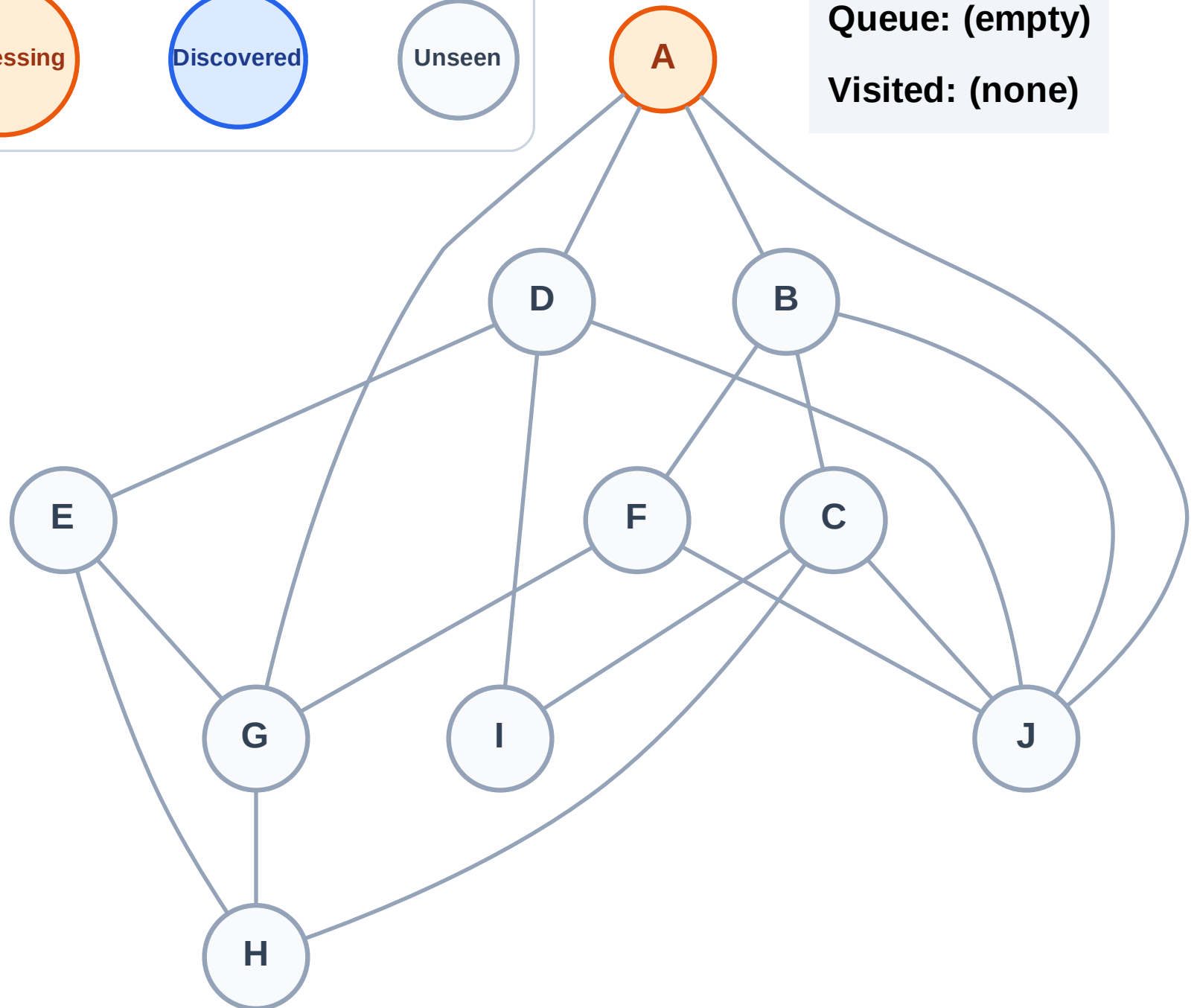
Step 2: Process node A

## Legend



Queue: (empty)

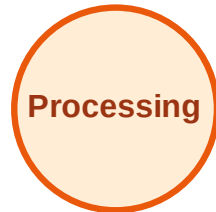
Visited: (none)



# BFS Traversal

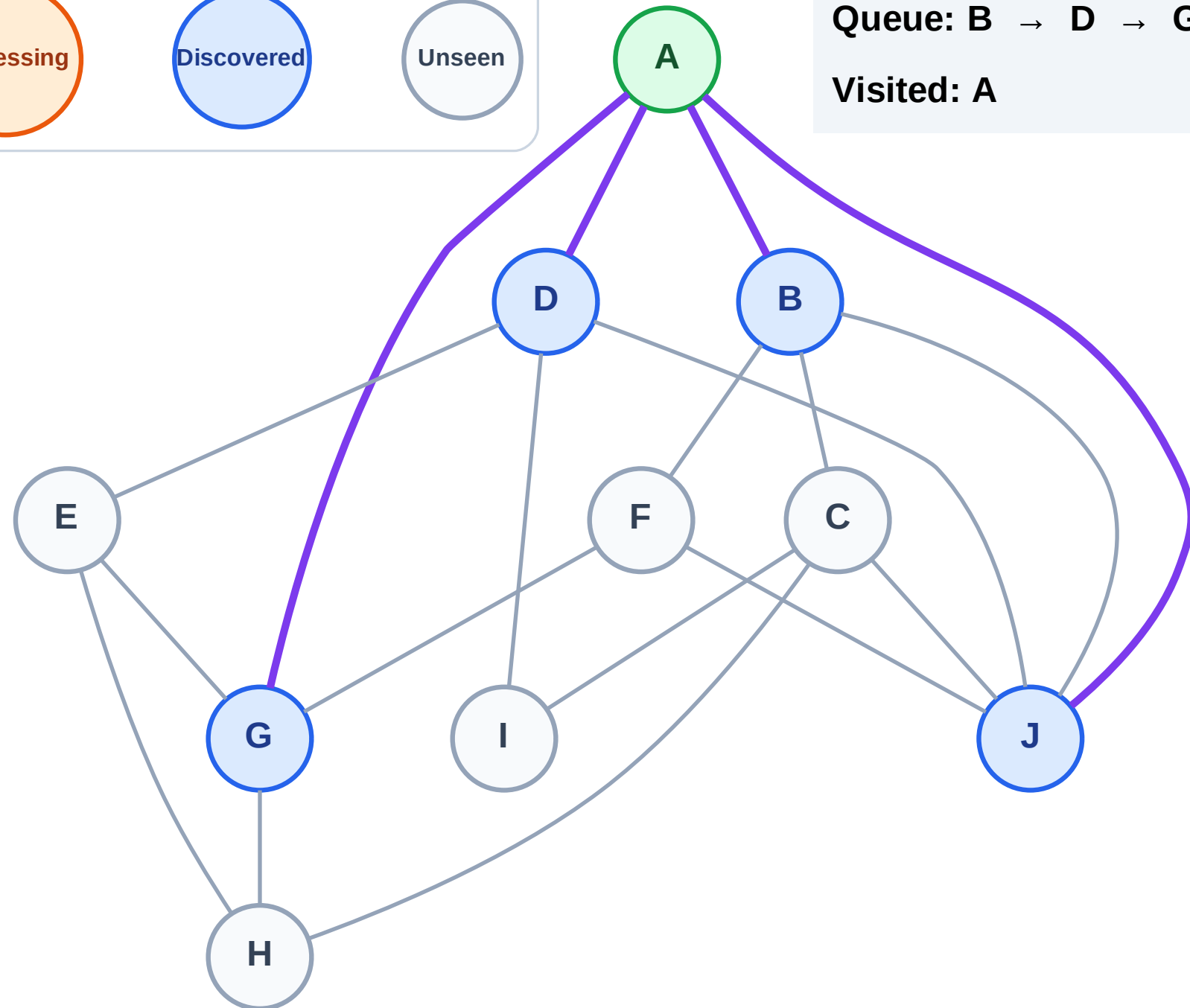
Step 3: Discover B, D, G, J from A

## Legend



Queue: B → D → G → J

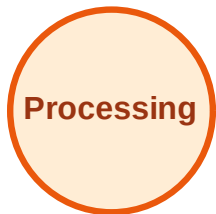
Visited: A



# BFS Traversal

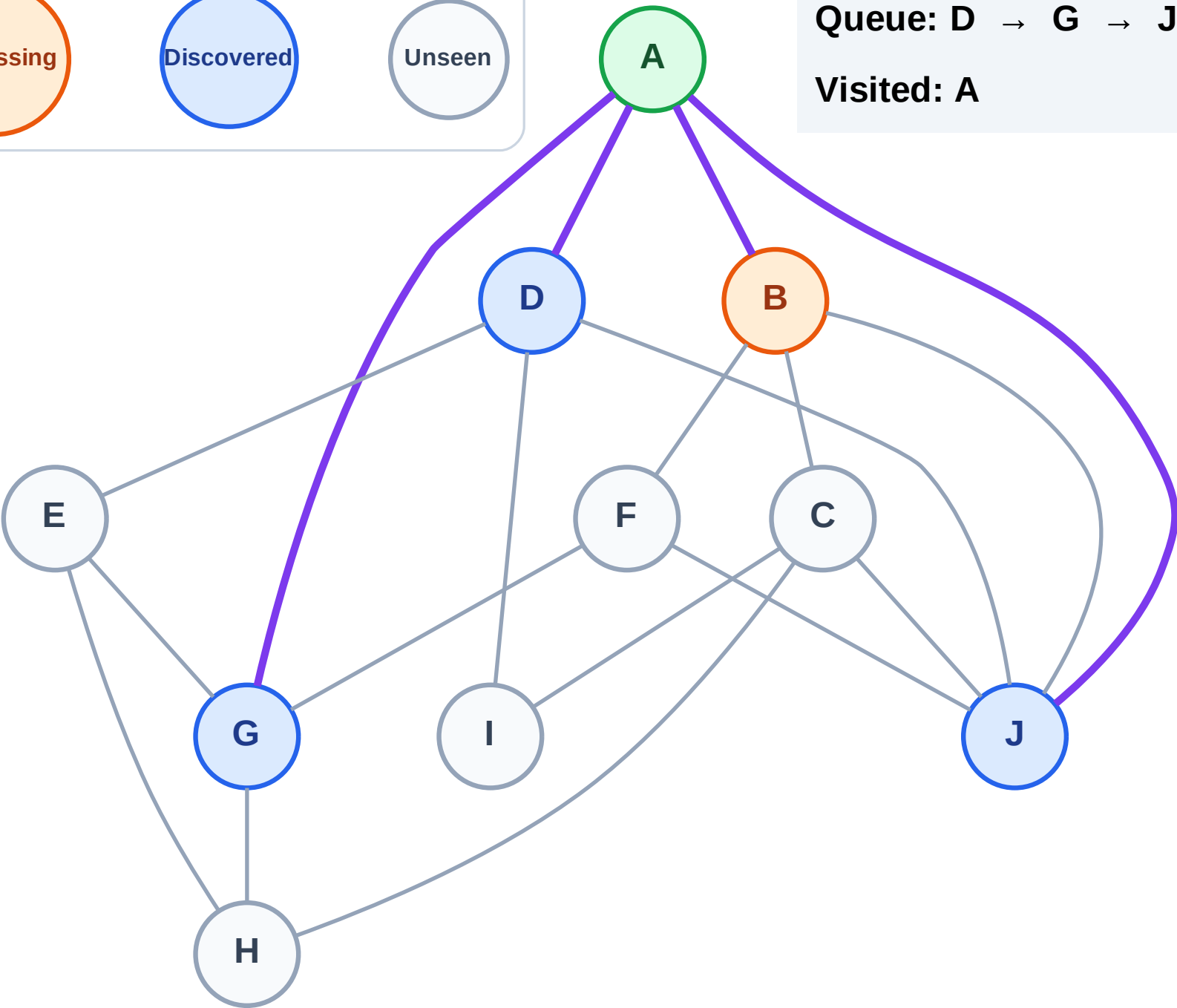
Step 4: Process node B

## Legend



Queue: D → G → J

Visited: A



# BFS Traversal

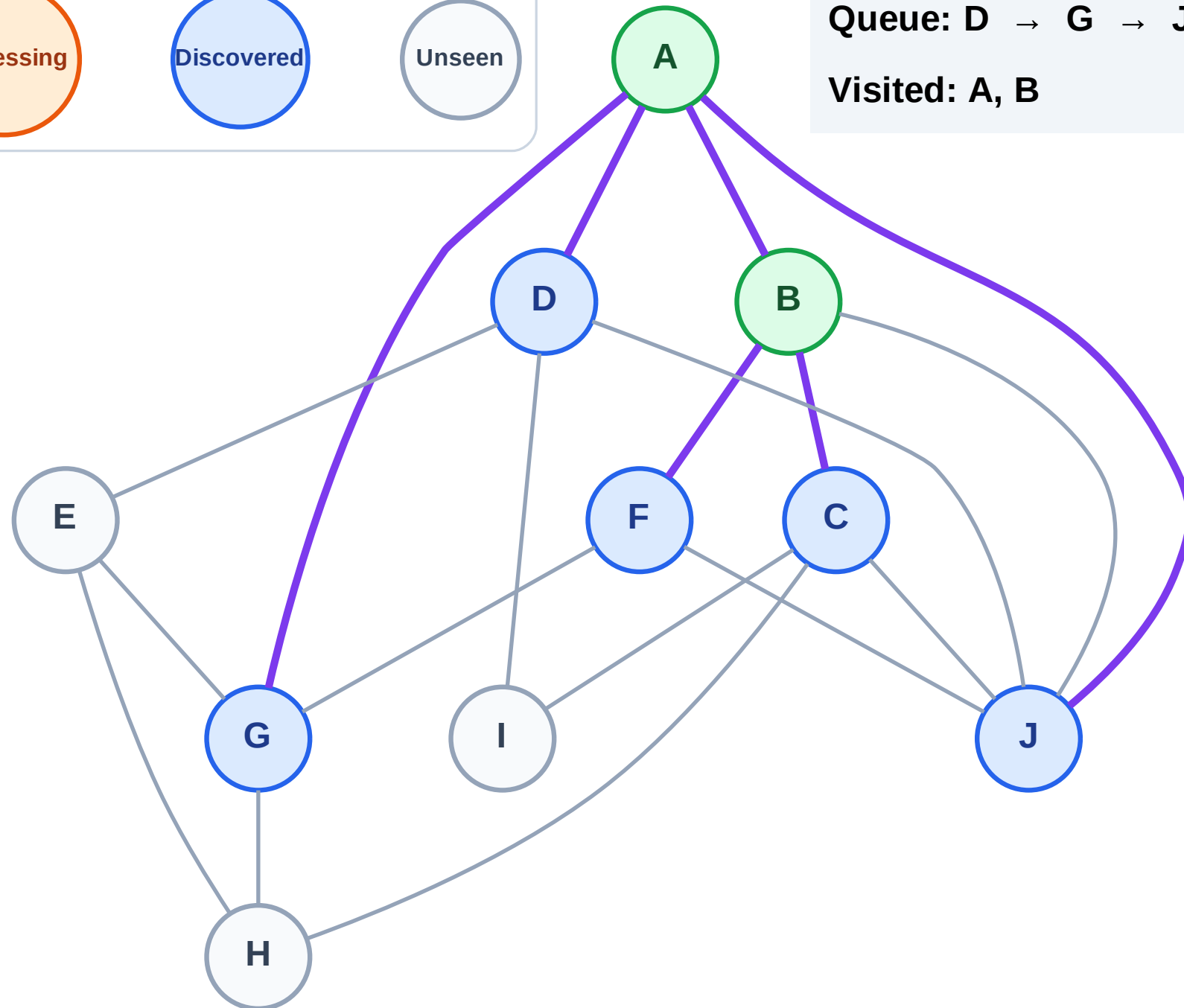
Step 5: Discover C, F from B

## Legend



Queue: D → G → J → C → F

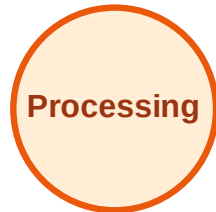
Visited: A, B



# BFS Traversal

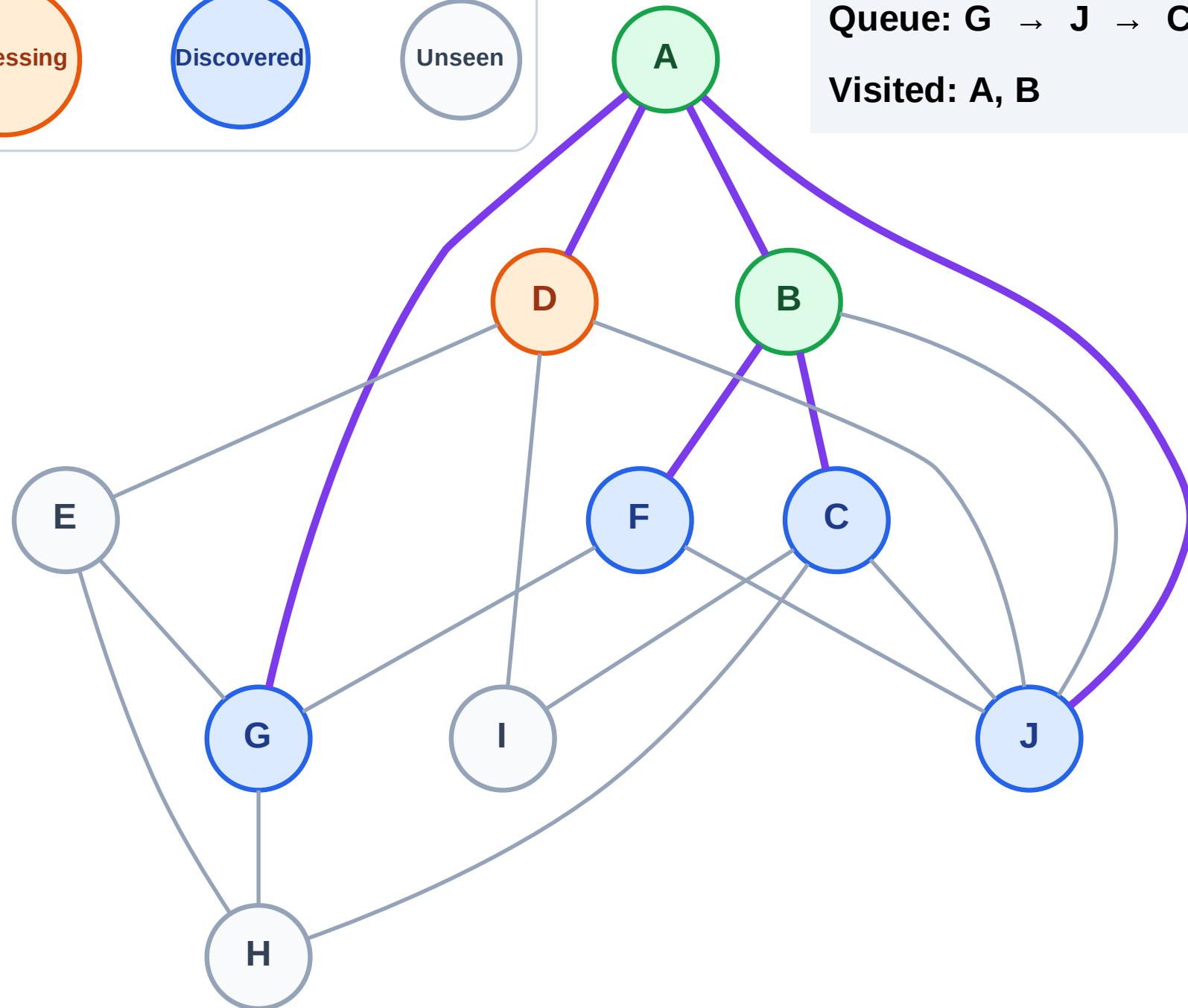
Step 6: Process node D

## Legend



Queue: G → J → C → F

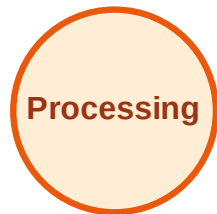
Visited: A, B



# BFS Traversal

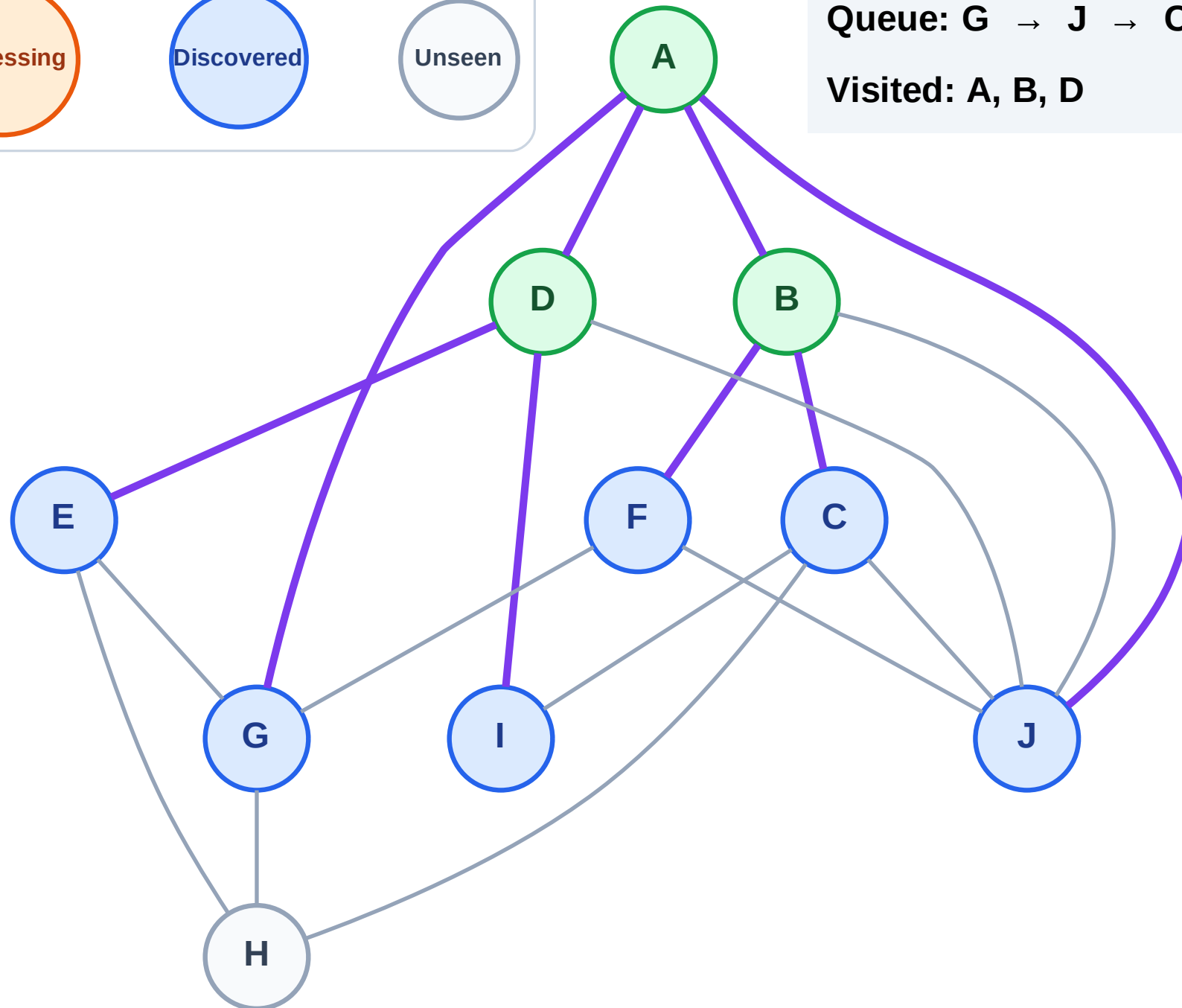
Step 7: Discover E, I from D

## Legend



Queue: G → J → C → F → E → I

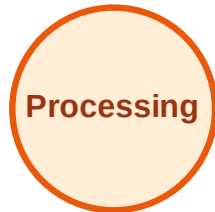
Visited: A, B, D



# BFS Traversal

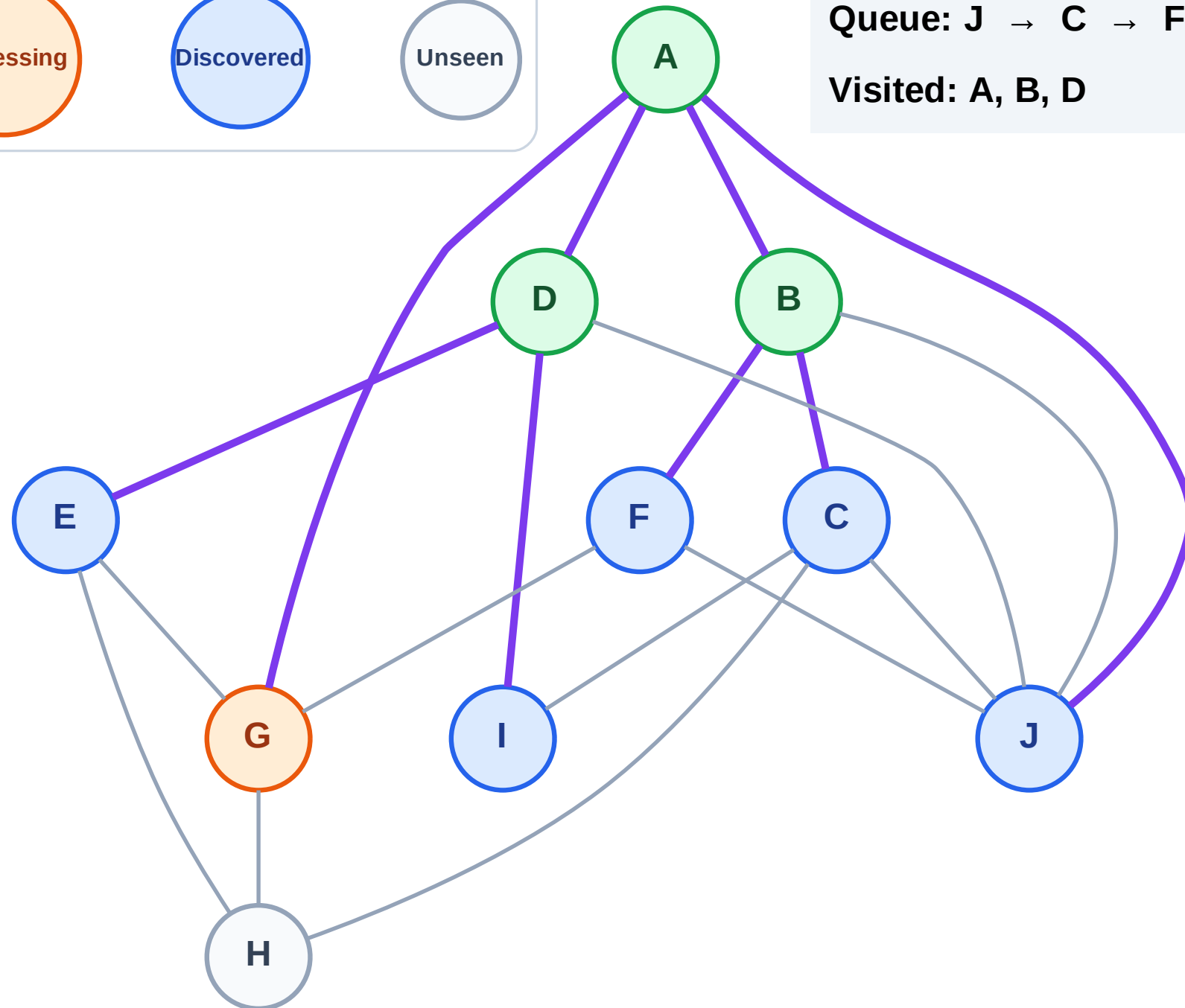
Step 8: Process node G

## Legend



Queue: J → C → F → E → I

Visited: A, B, D

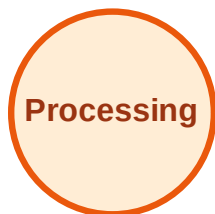




# BFS Traversal

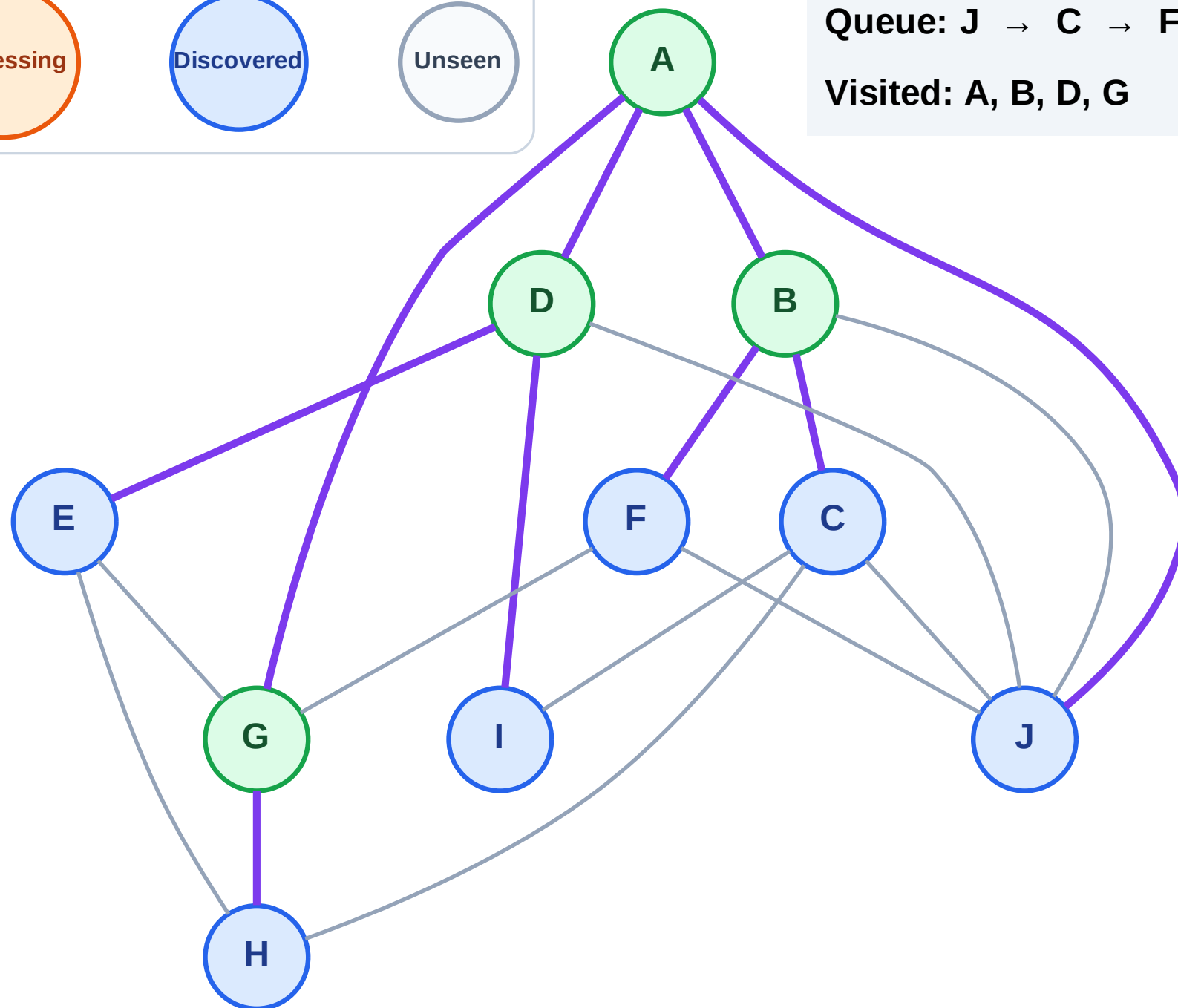
Step 9: Discover H from G

## Legend



Queue: J → C → F → E → I → H

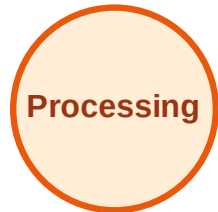
Visited: A, B, D, G



# BFS Traversal

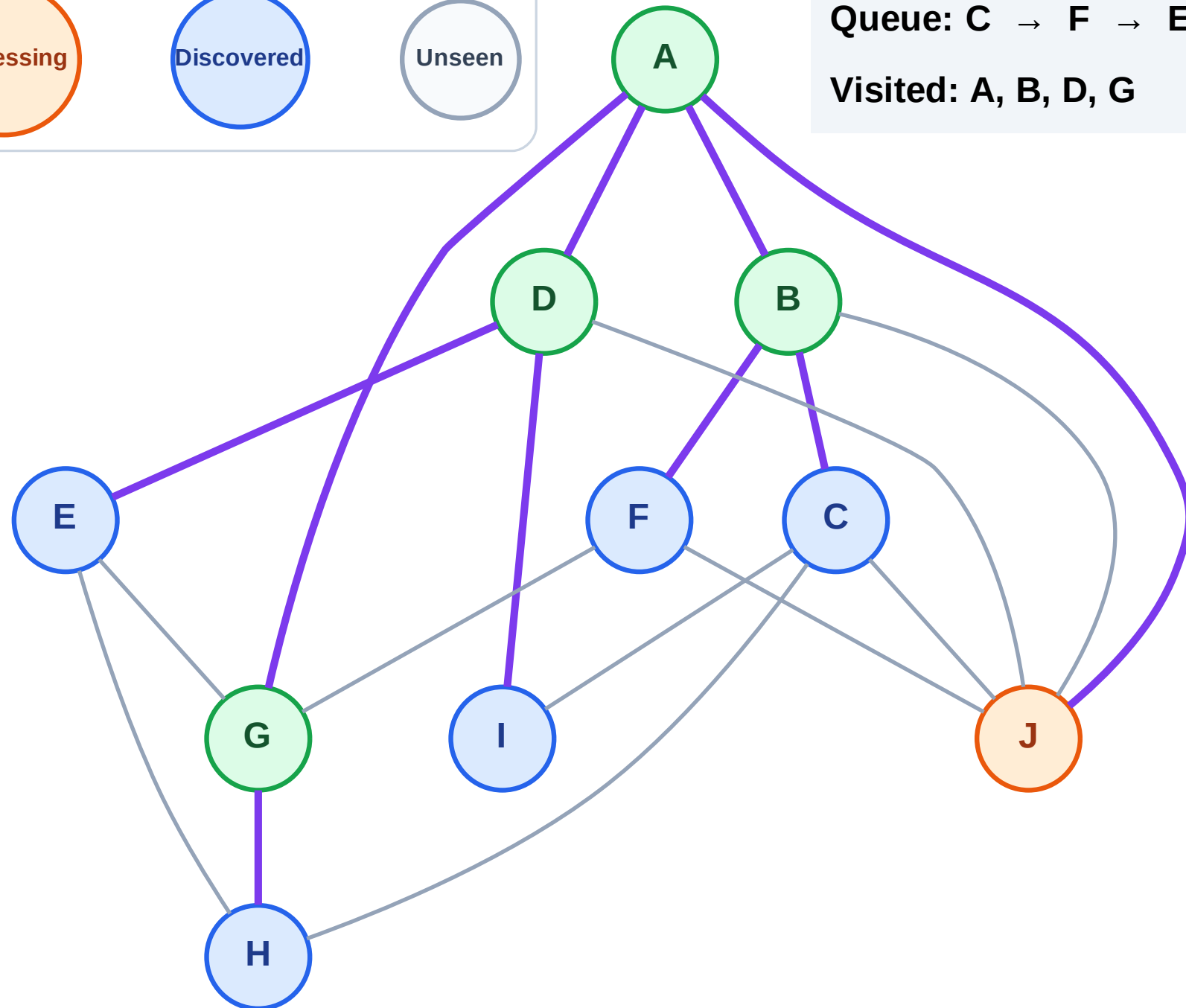
Step 10: Process node J

## Legend



Queue: C → F → E → I → H

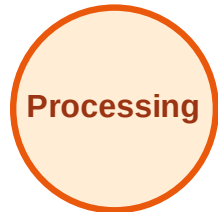
Visited: A, B, D, G



# BFS Traversal

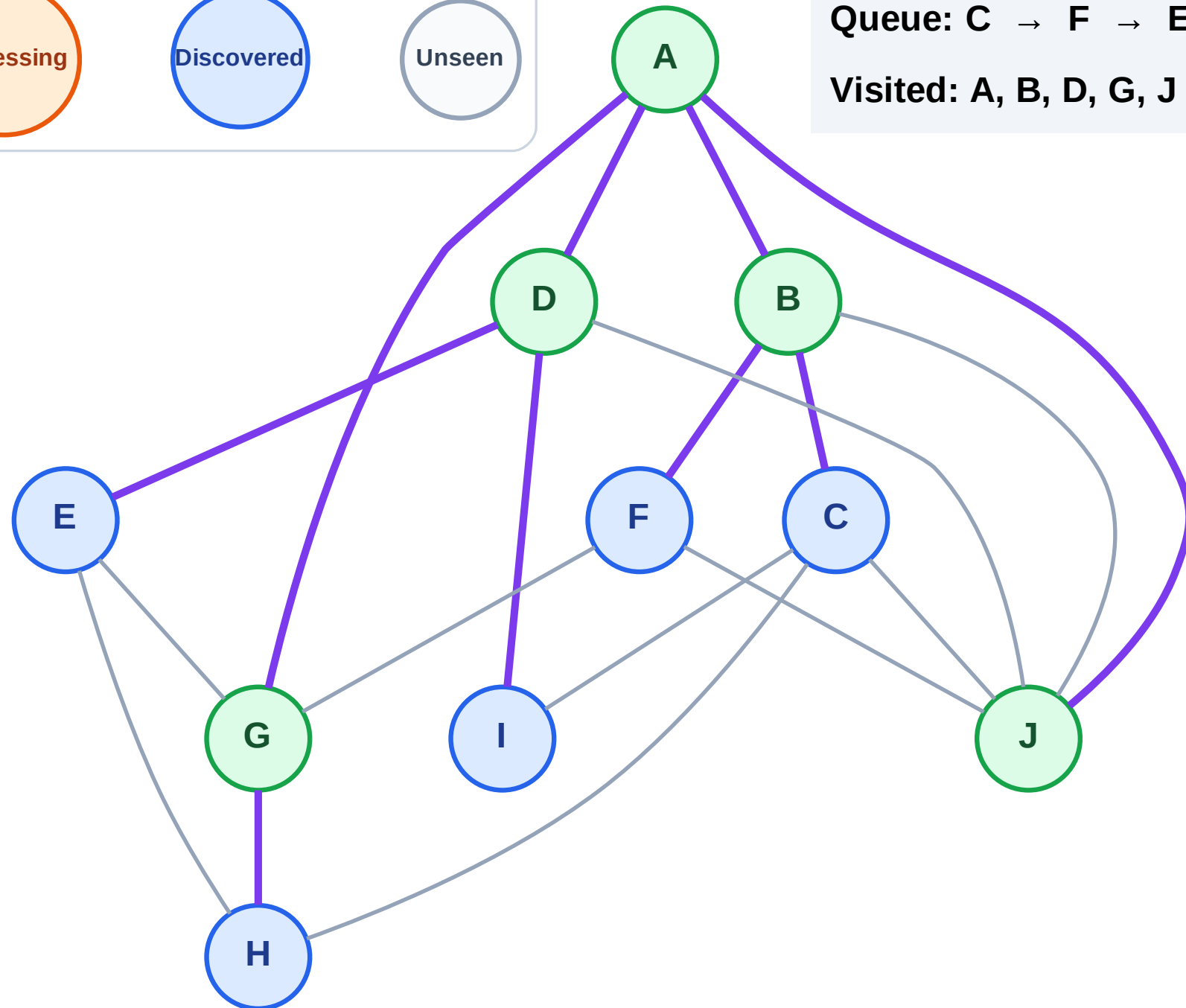
Step 11: No new neighbors from J

## Legend



Queue: C → F → E → I → H

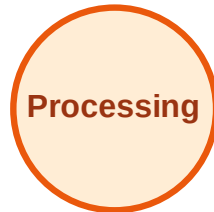
Visited: A, B, D, G, J



# BFS Traversal

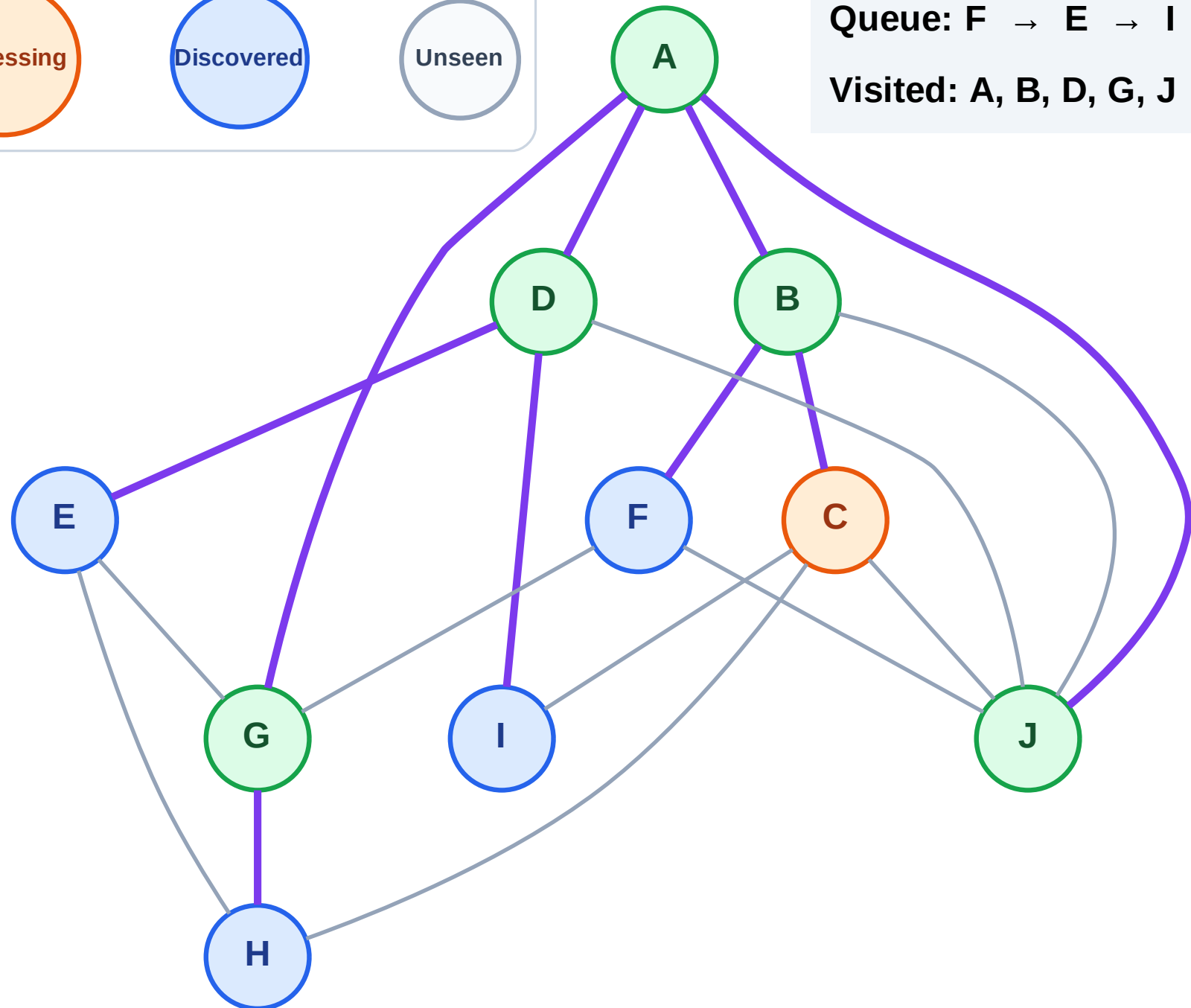
Step 12: Process node C

## Legend



Queue: F → E → I → H

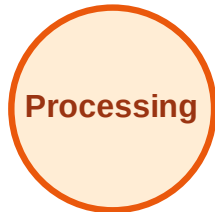
Visited: A, B, D, G, J



# BFS Traversal

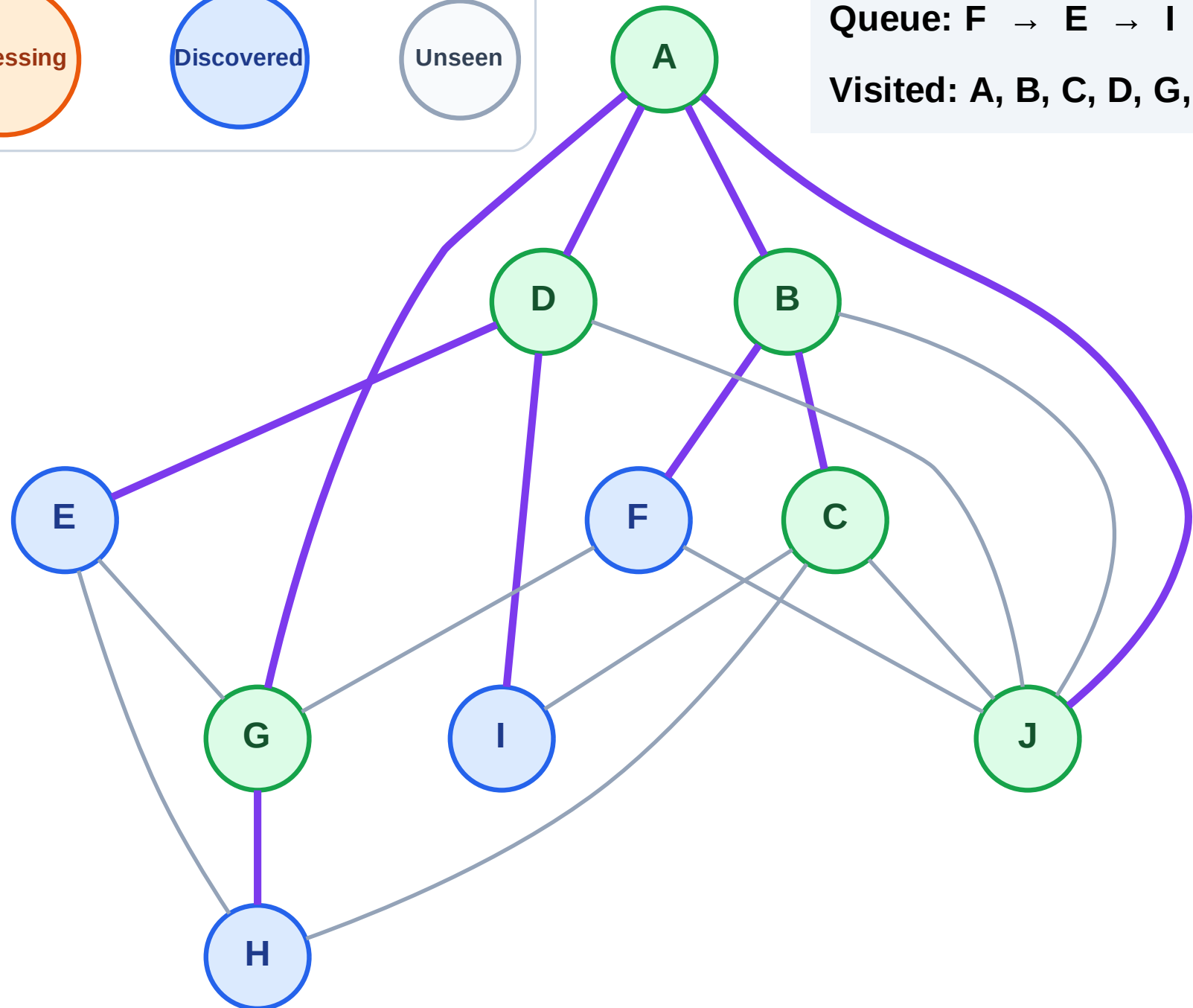
Step 13: No new neighbors from C

## Legend



Queue: F → E → I → H

Visited: A, B, C, D, G, J



# BFS Traversal

Step 14: Process node F

Legend

Complete

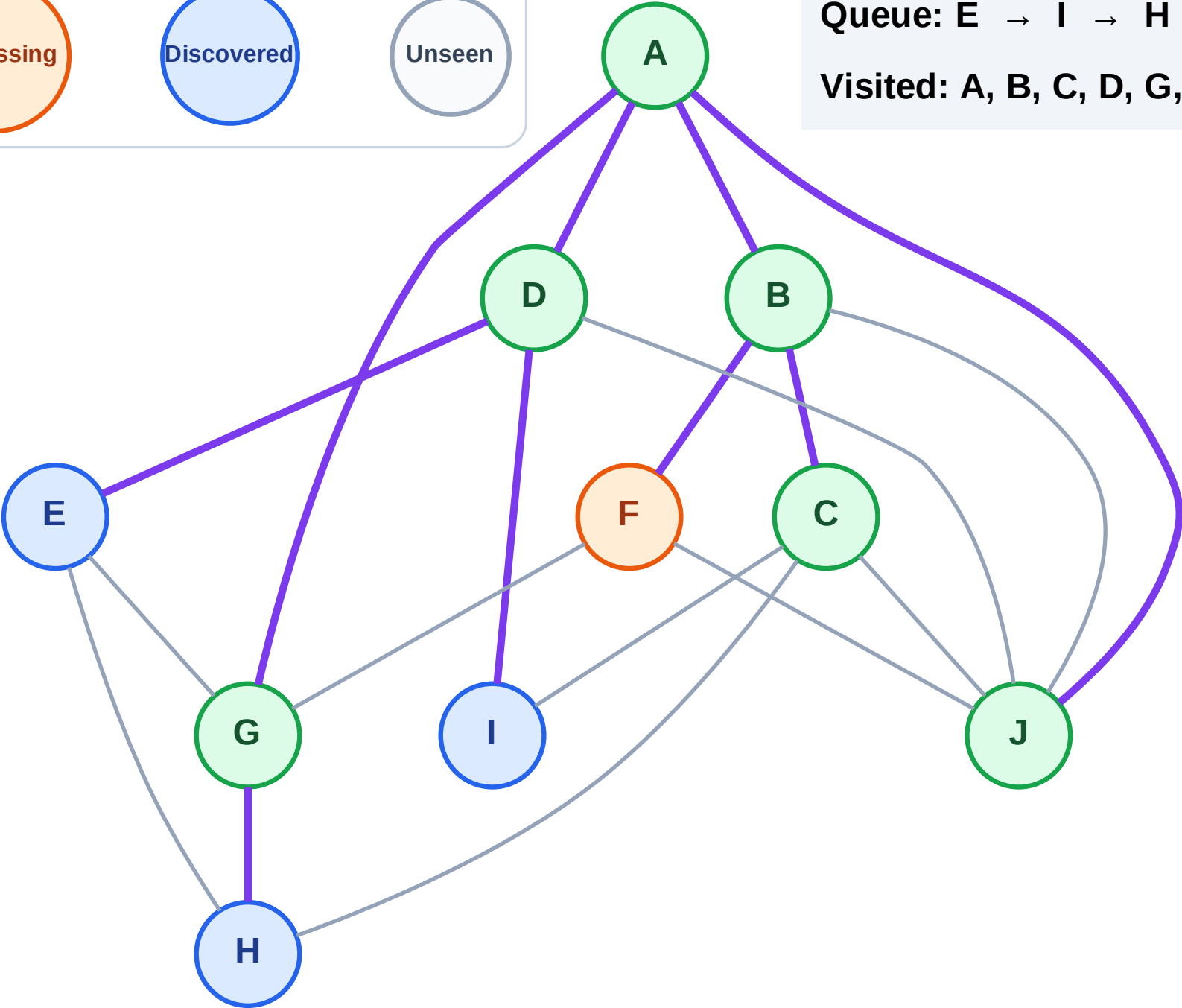
Processing

Discovered

Unseen

Queue: E → I → H

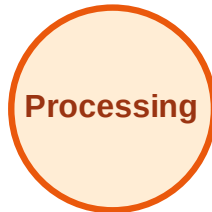
Visited: A, B, C, D, G, J



# BFS Traversal

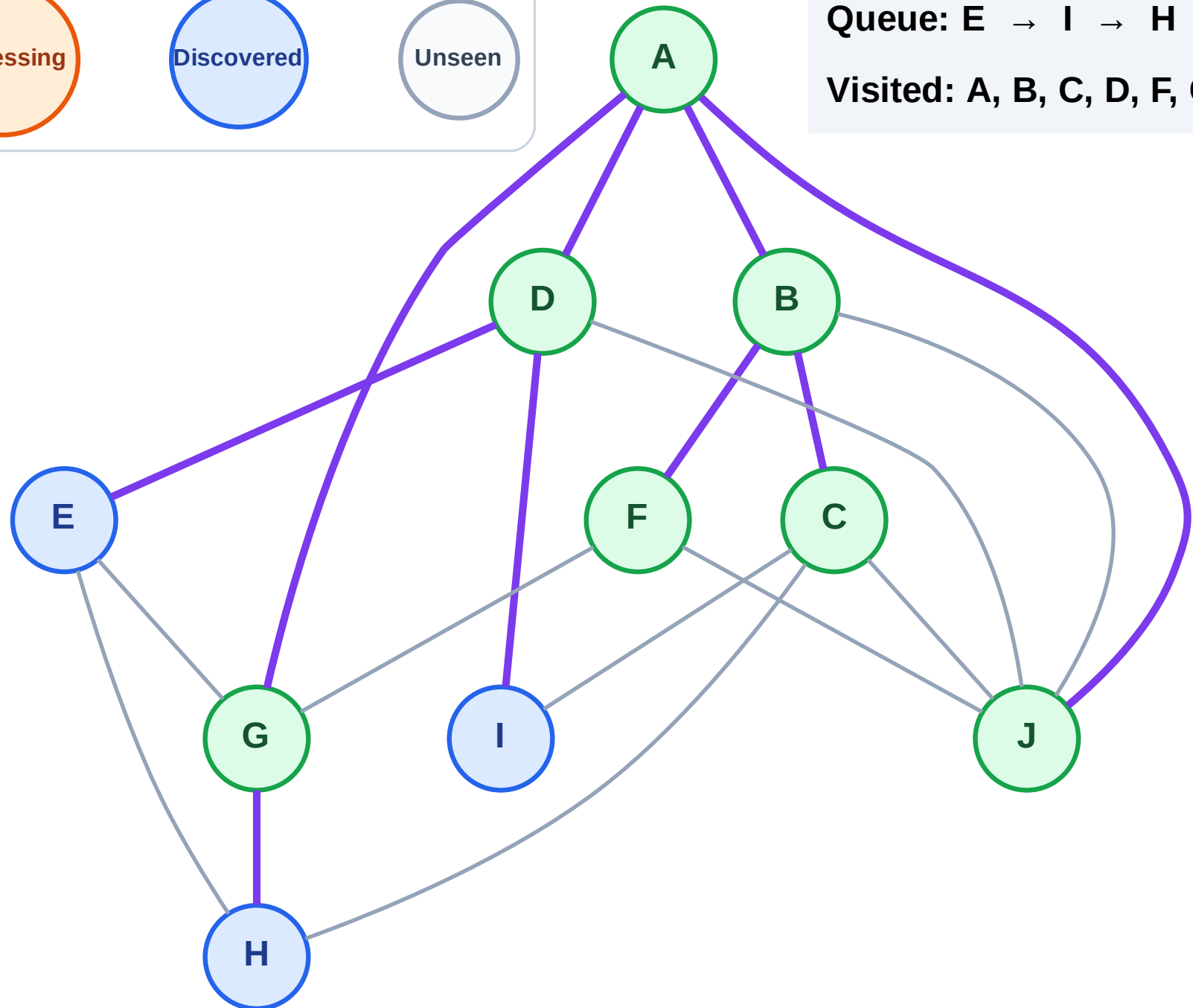
Step 15: No new neighbors from F

## Legend



Queue: E → I → H

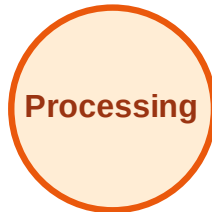
Visited: A, B, C, D, F, G, J



# BFS Traversal

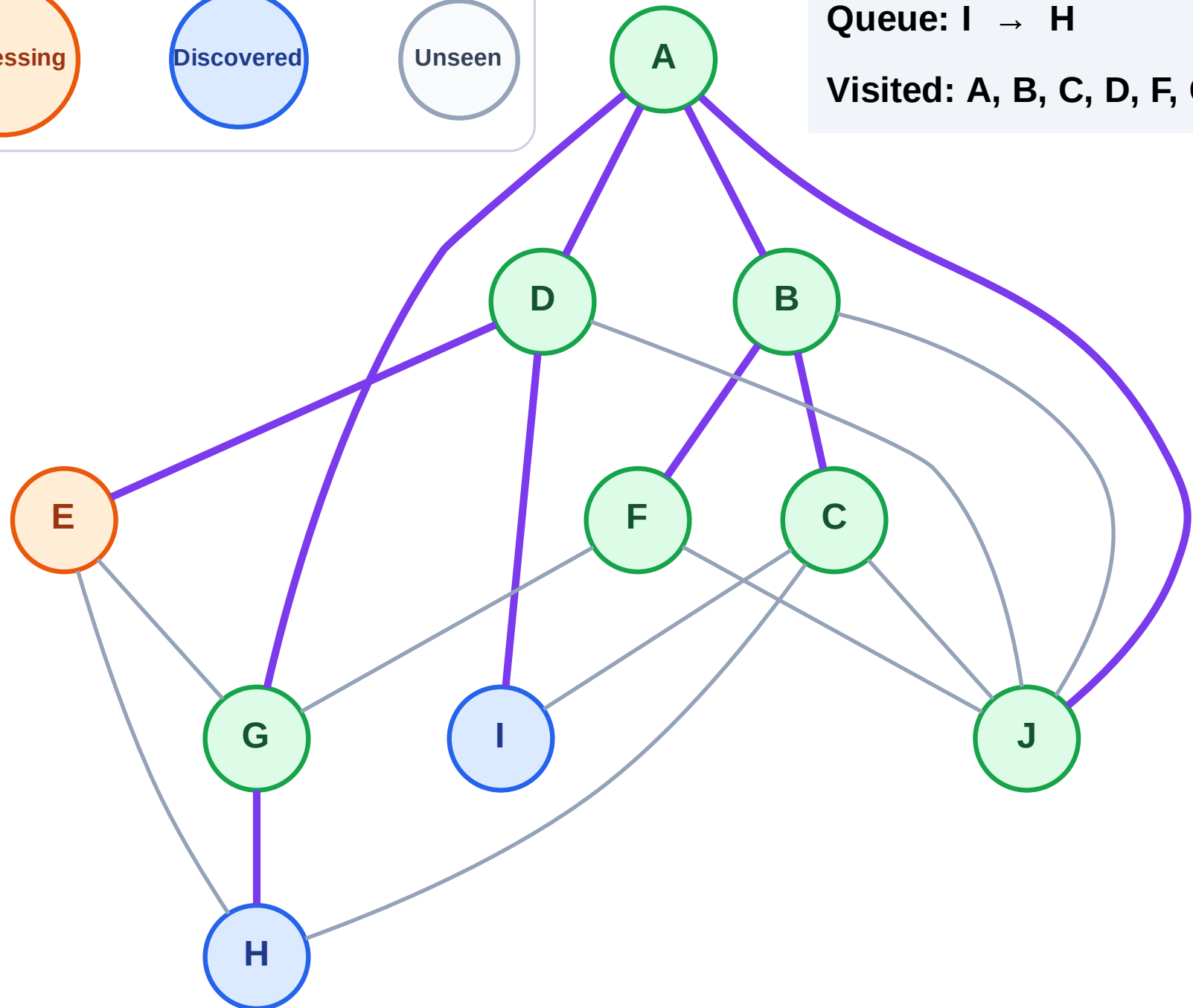
Step 16: Process node E

## Legend



Queue: I → H

Visited: A, B, C, D, F, G, J

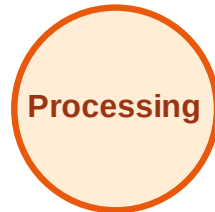




# BFS Traversal

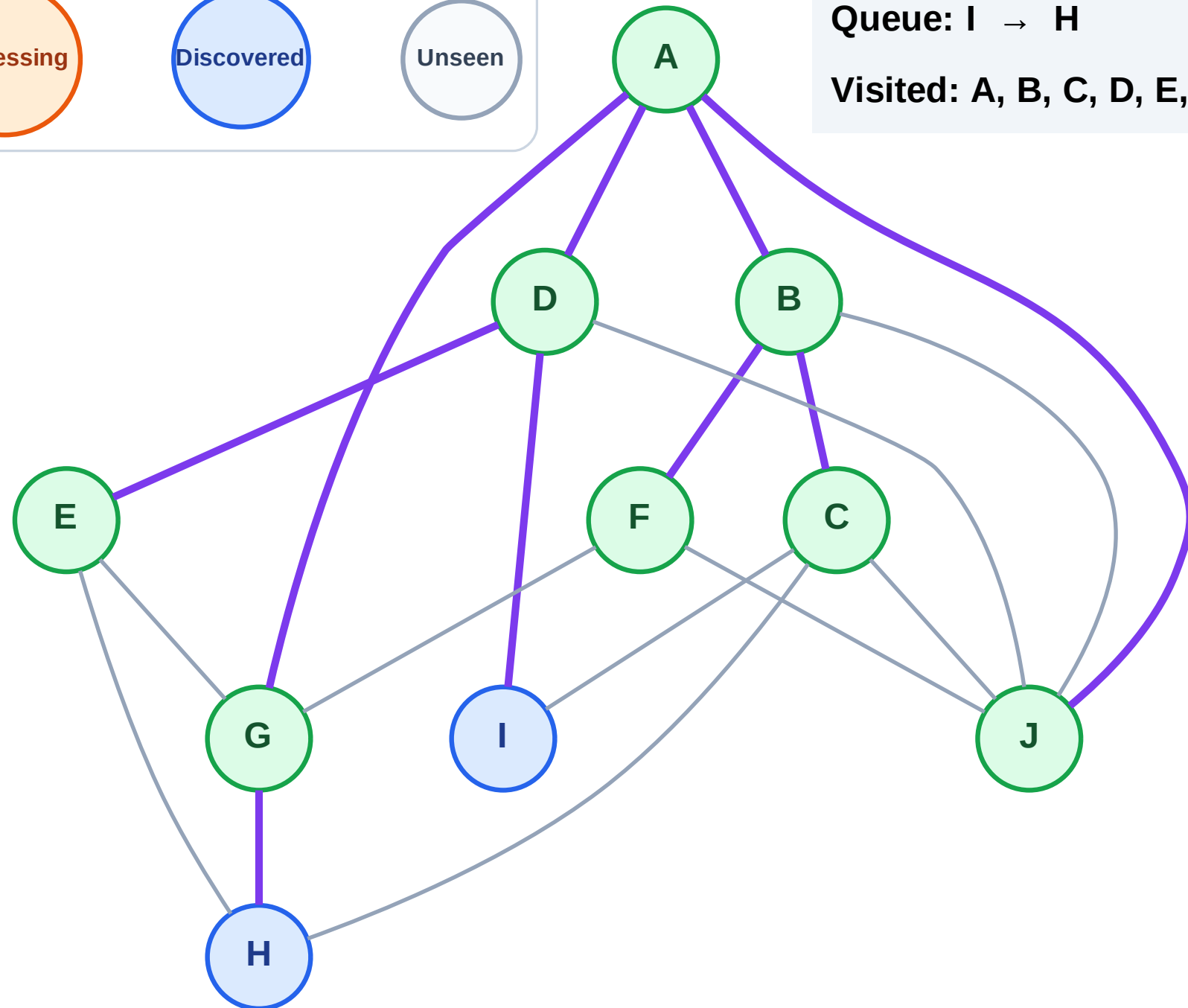
Step 17: No new neighbors from E

## Legend



Queue: I → H

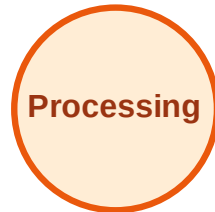
Visited: A, B, C, D, E, F, G, J



# BFS Traversal

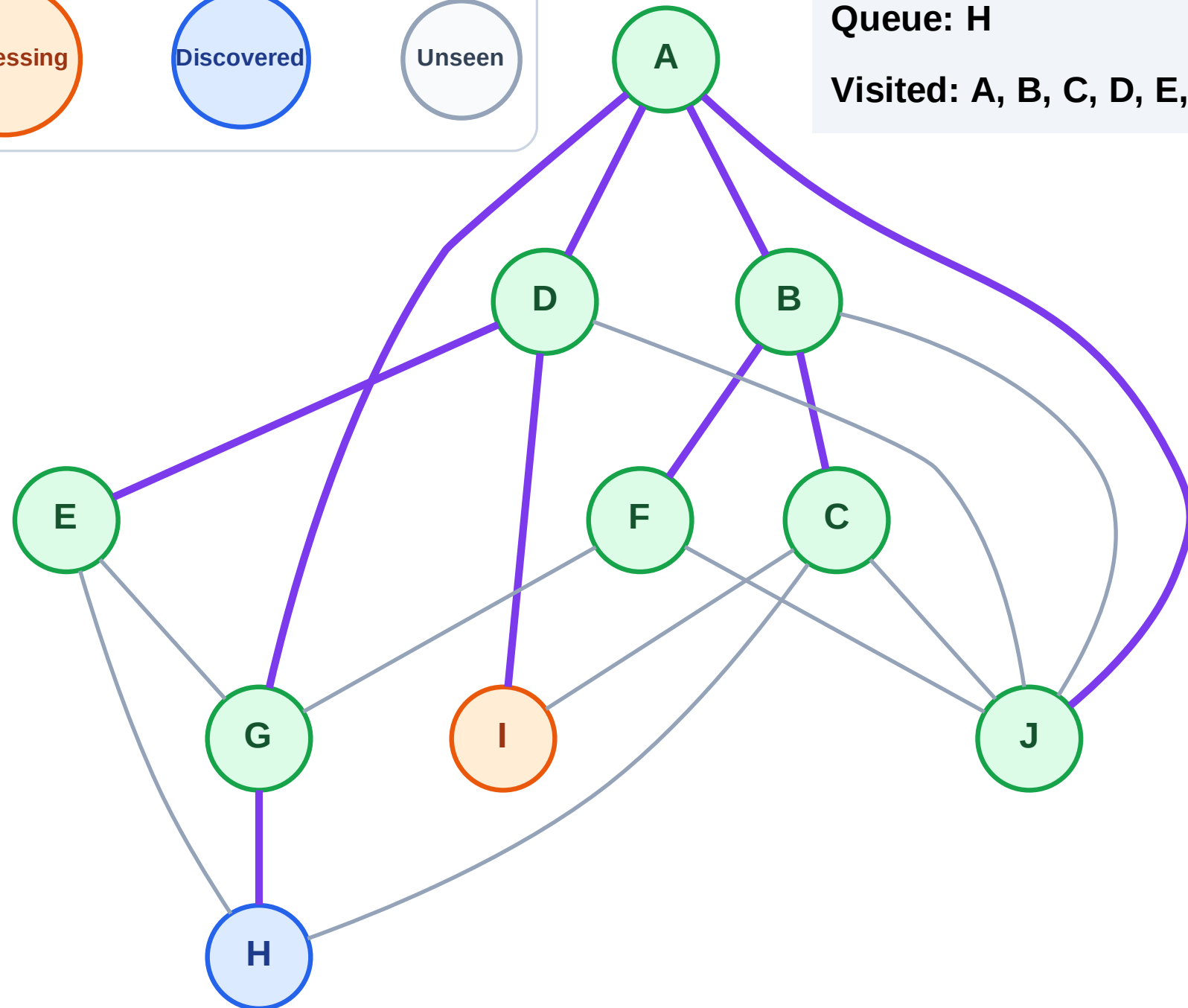
Step 18: Process node I

## Legend



Queue: H

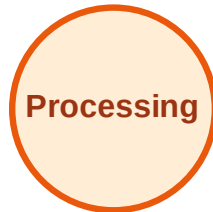
Visited: A, B, C, D, E, F, G, J



# BFS Traversal

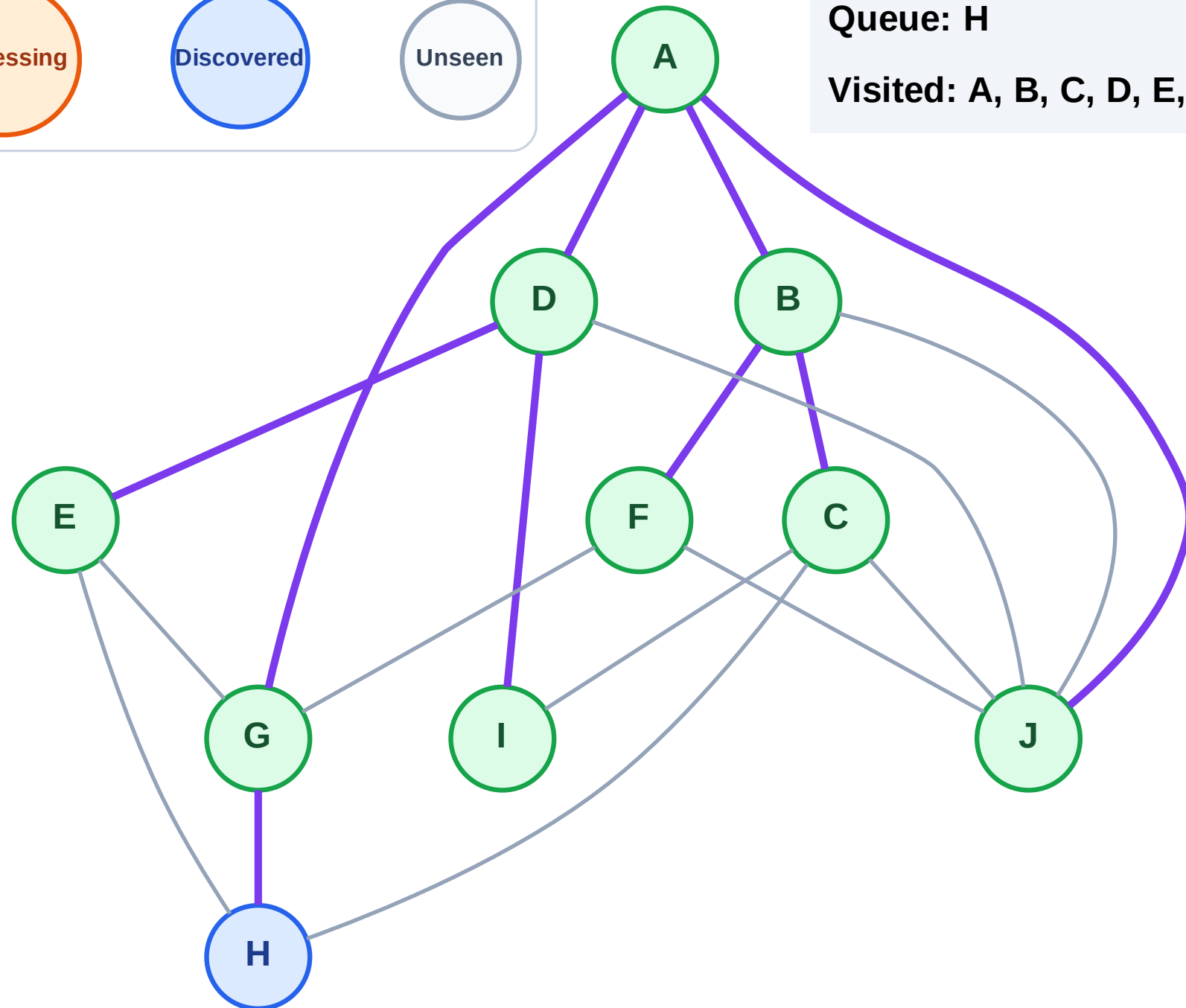
Step 19: No new neighbors from I

## Legend



Queue: H

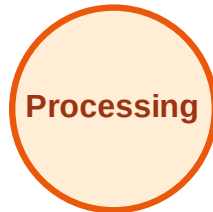
Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

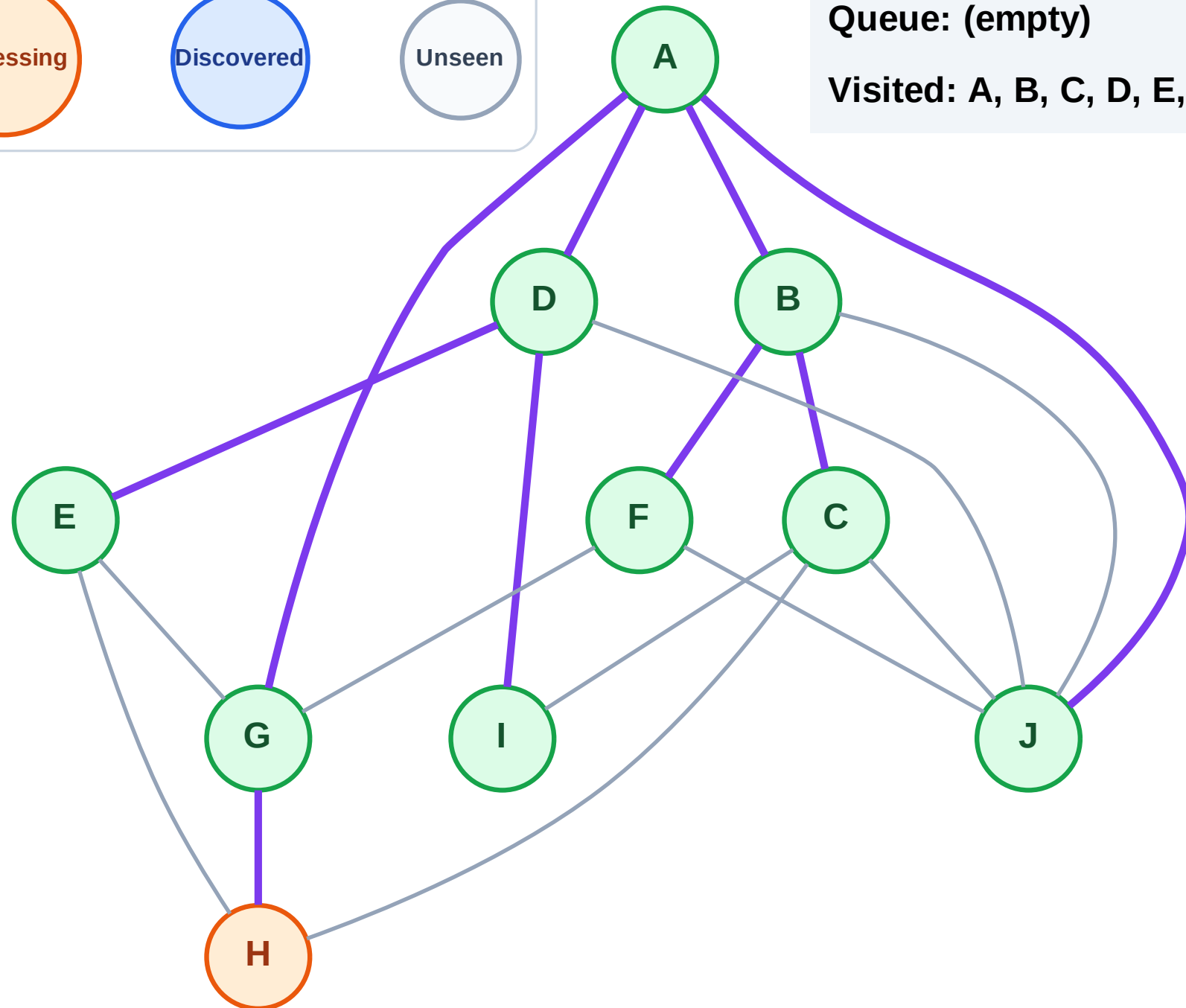
Step 20: Process node H

## Legend



Queue: (empty)

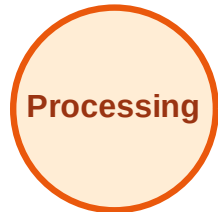
Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

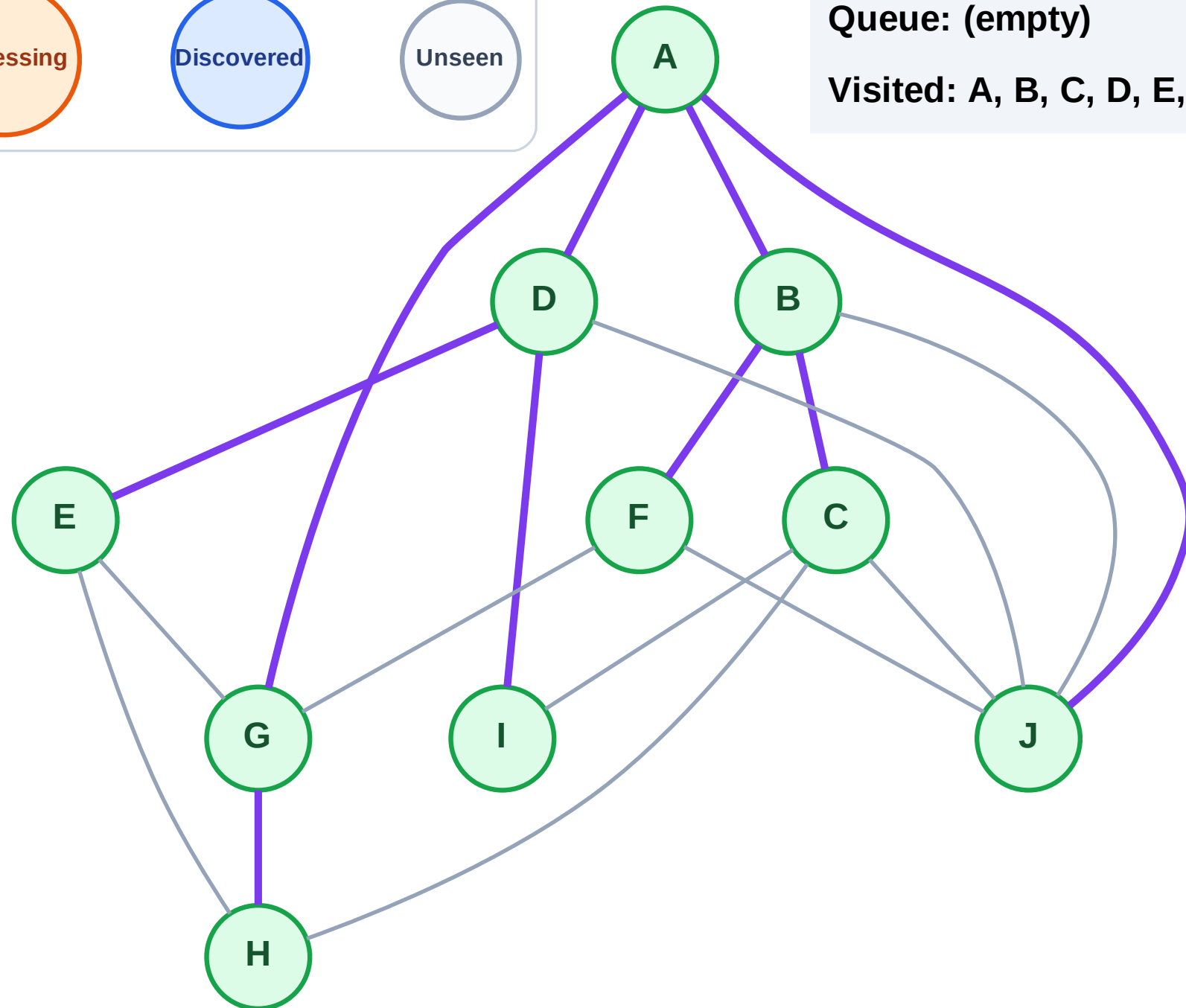
Step 21: No new neighbors from H

## Legend



Queue: (empty)

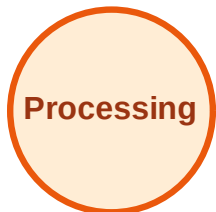
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

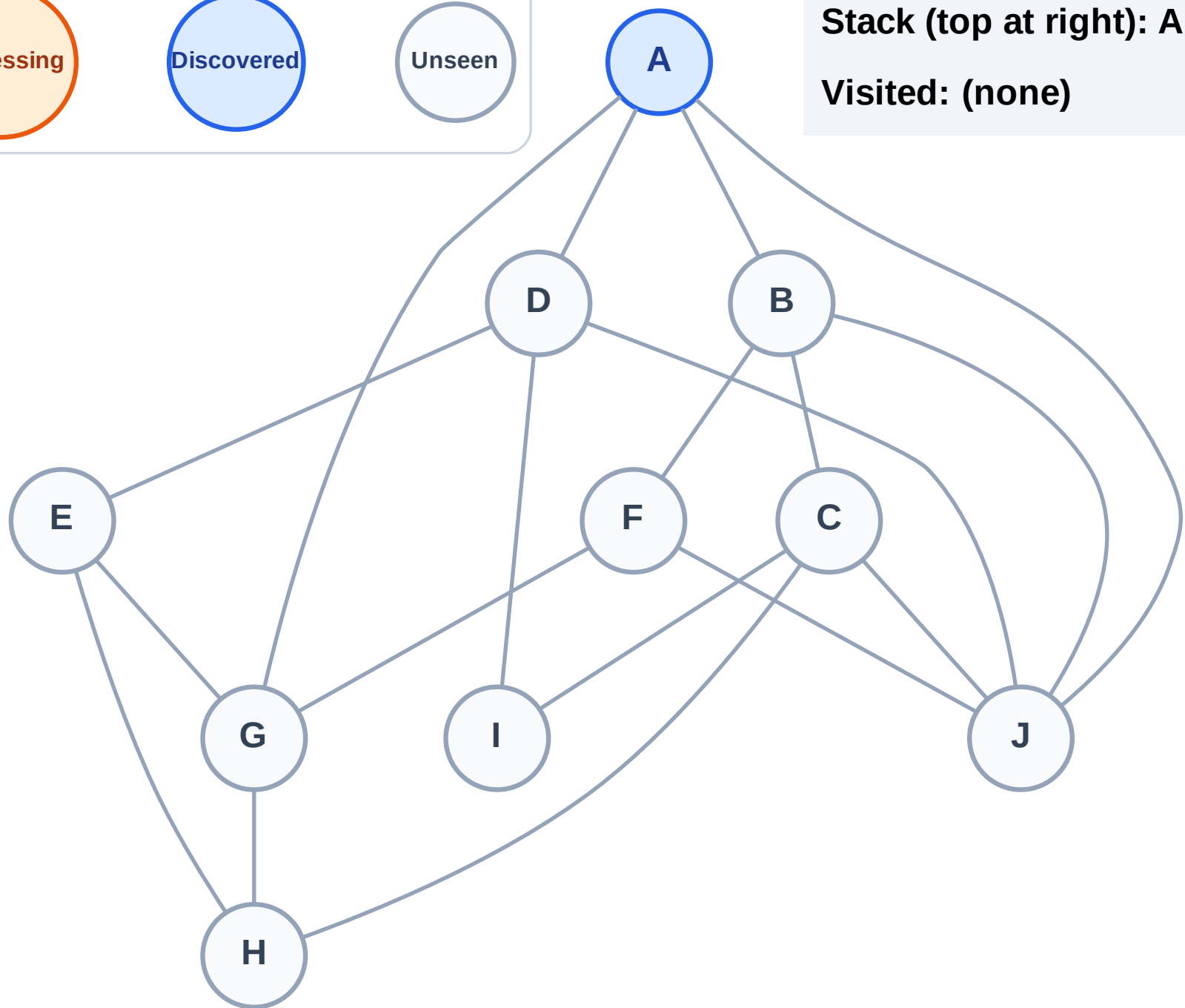
Step 1: Start at A

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

## Legend

Complete

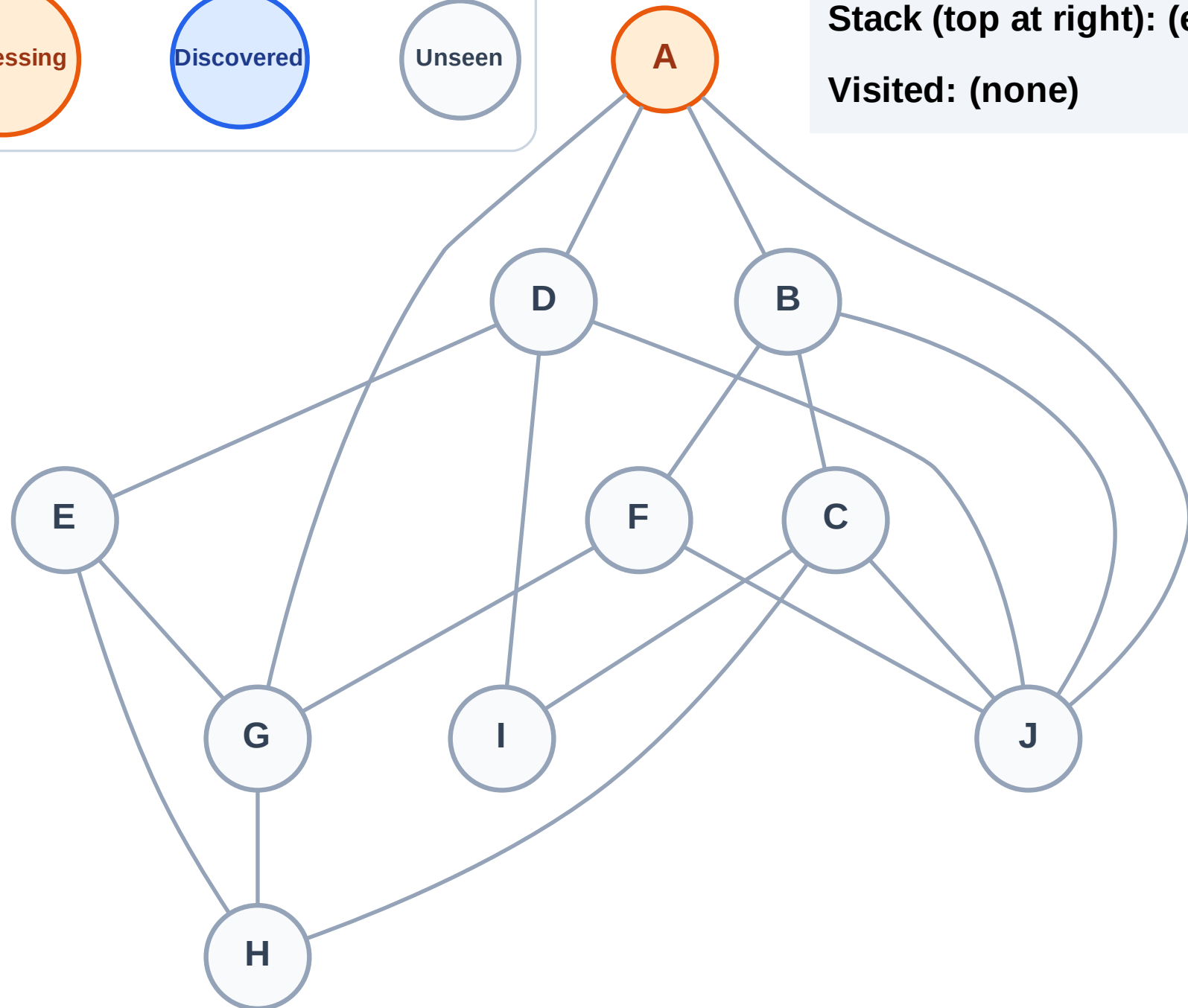
Processing

Discovered

Unseen

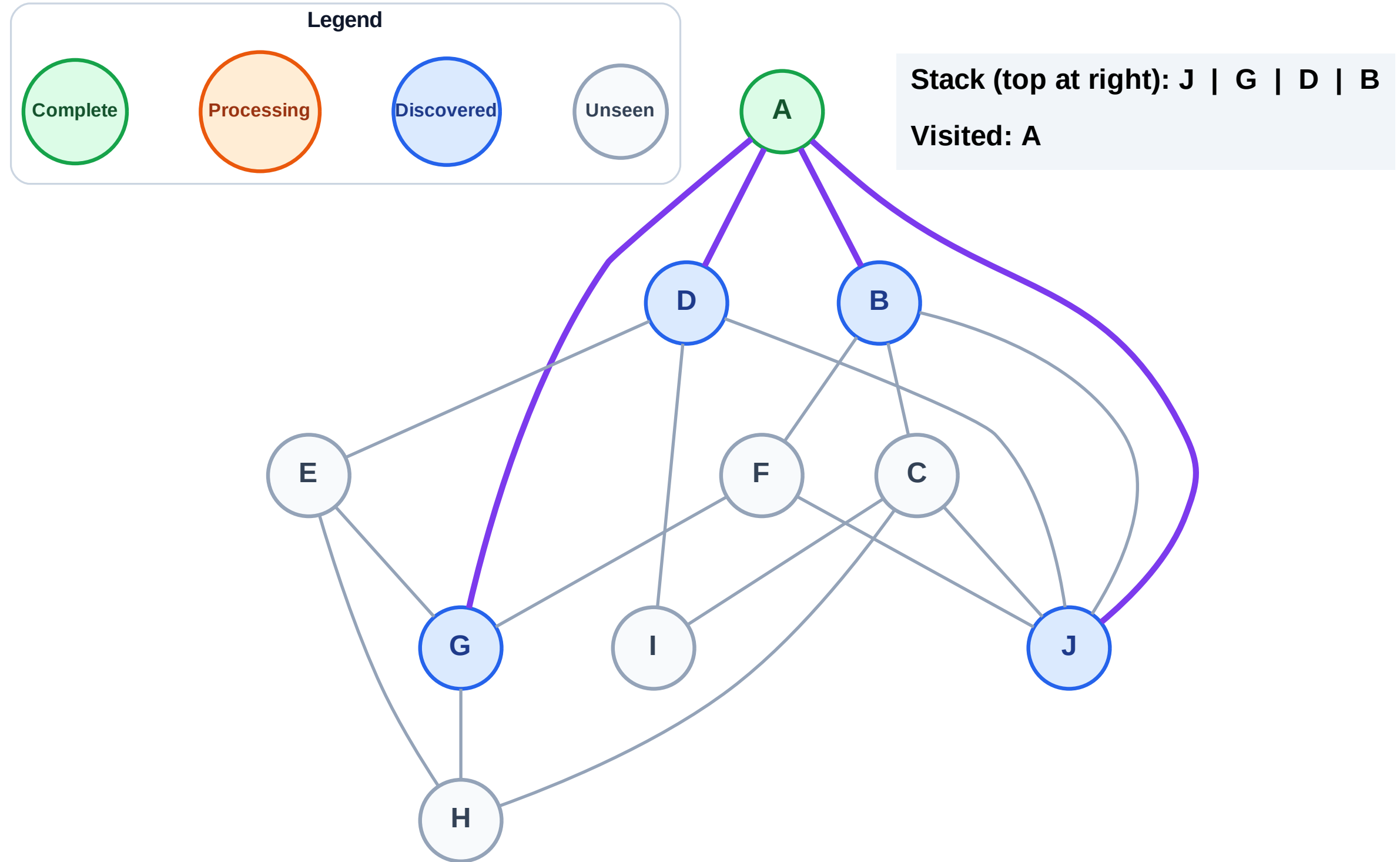
Stack (top at right): (empty)

Visited: (none)



### Step 3: Discover J, G, D, B from A

### Step 3: Discover J, G, D, B from A

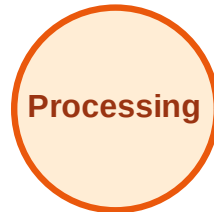




# DFS Traversal

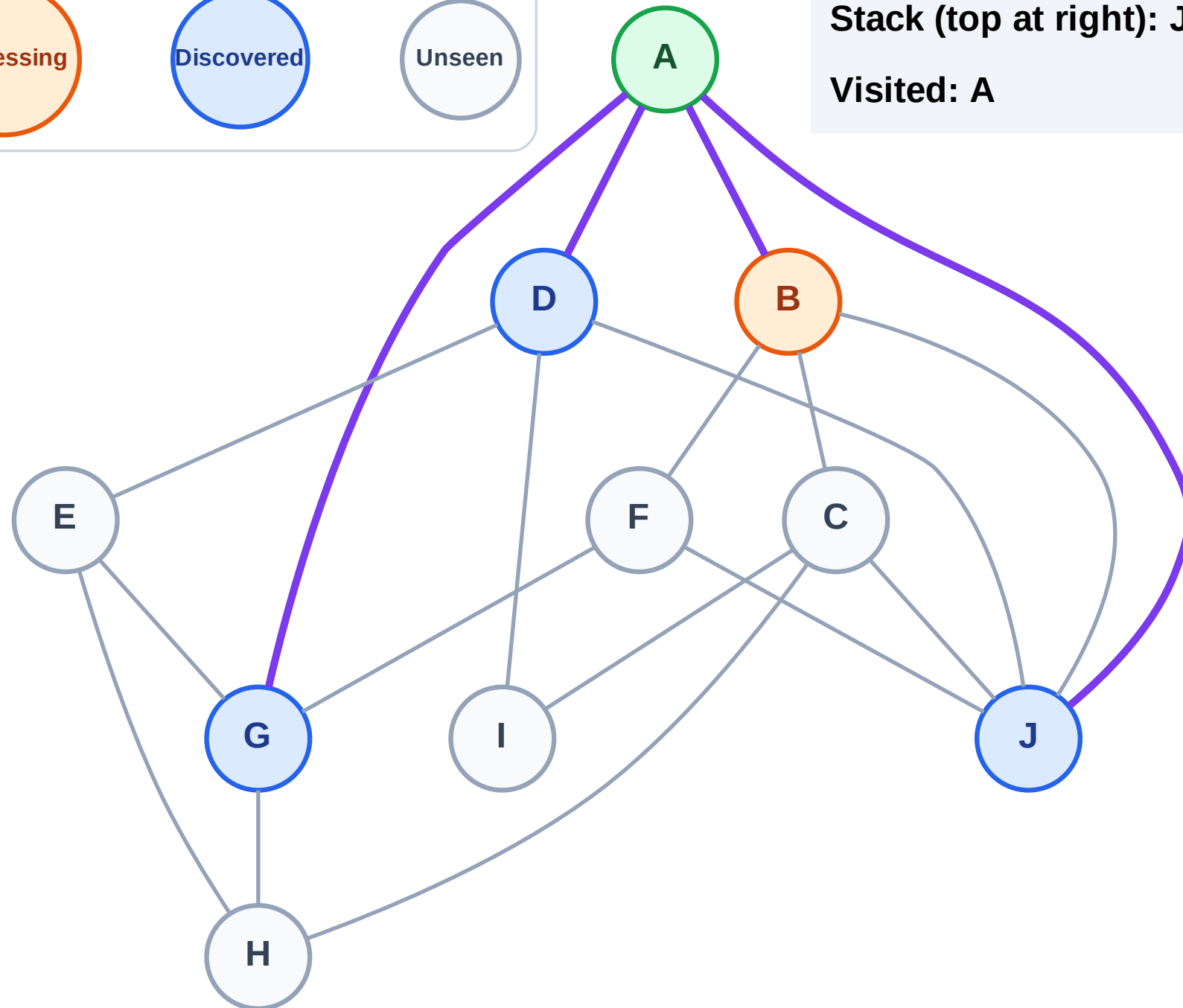
Step 4: Process node B

## Legend



Stack (top at right): J | G | D

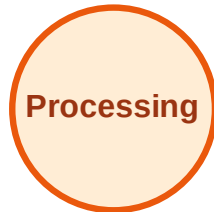
Visited: A



# DFS Traversal

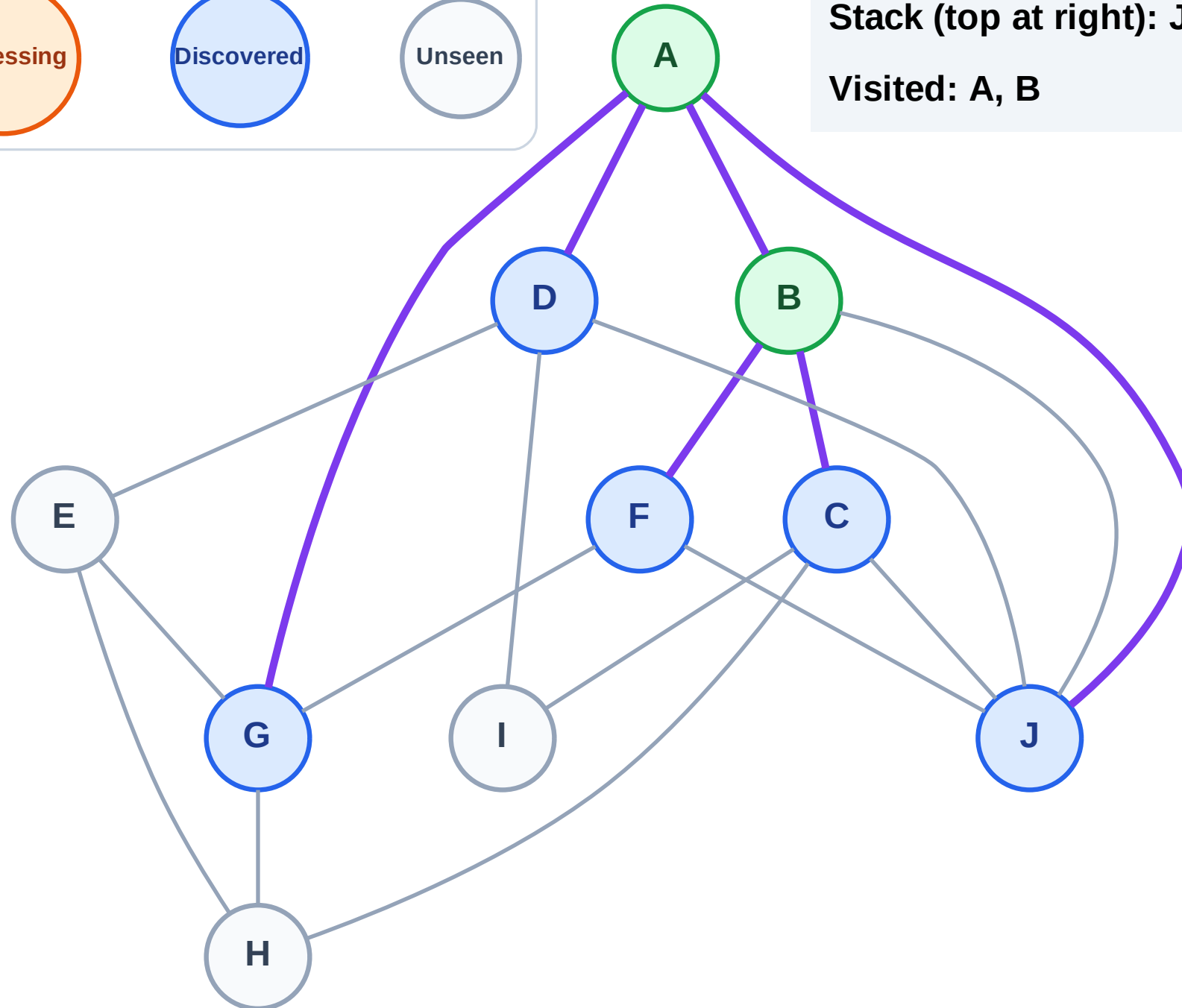
Step 5: Discover F, C from B

## Legend



Stack (top at right): J | G | D | F | C

Visited: A, B



# DFS Traversal

Step 6: Process node C

Legend

Complete

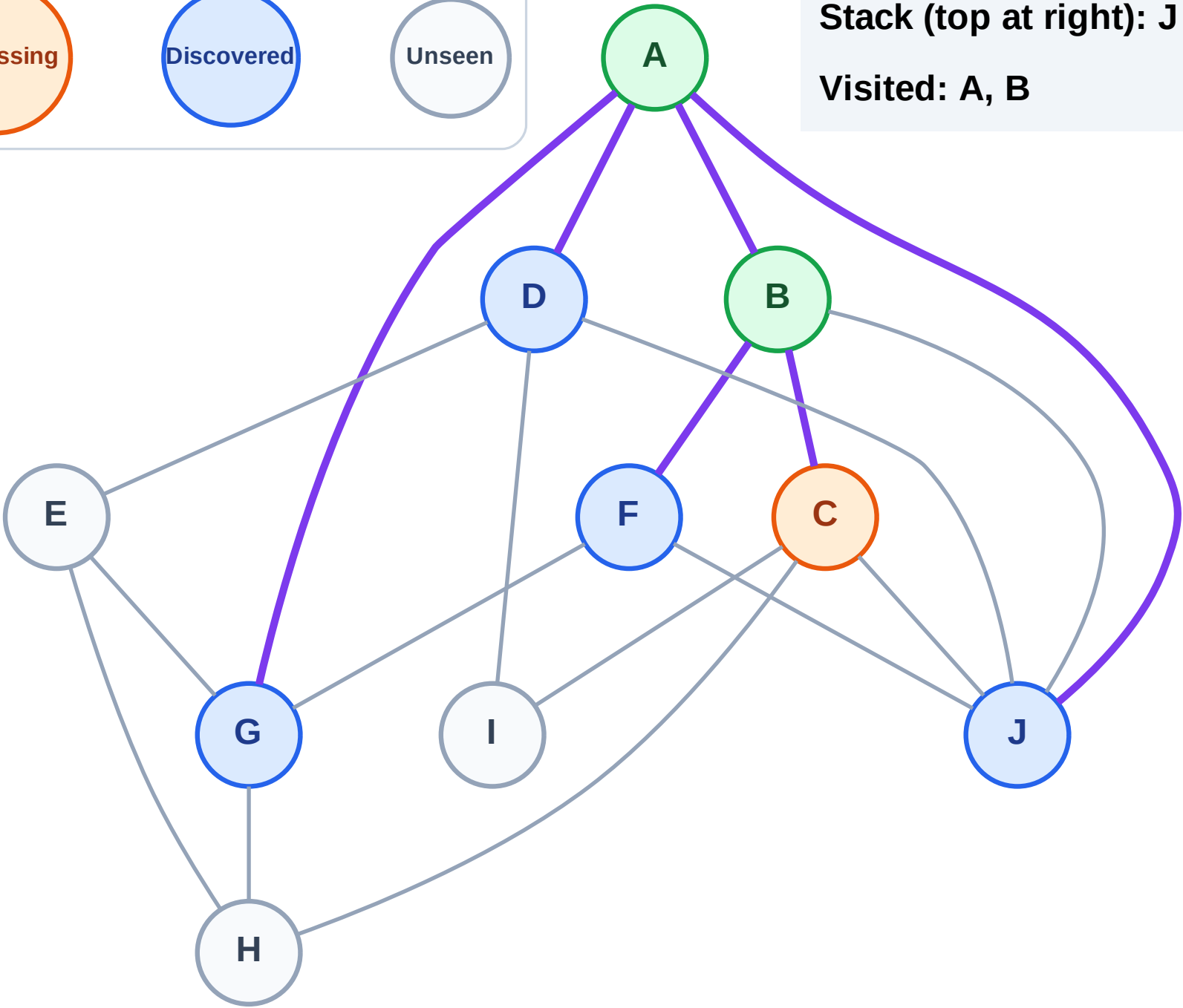
Processing

Discovered

Unseen

Stack (top at right): J | G | D | F

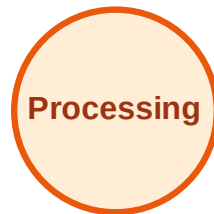
Visited: A, B



# DFS Traversal

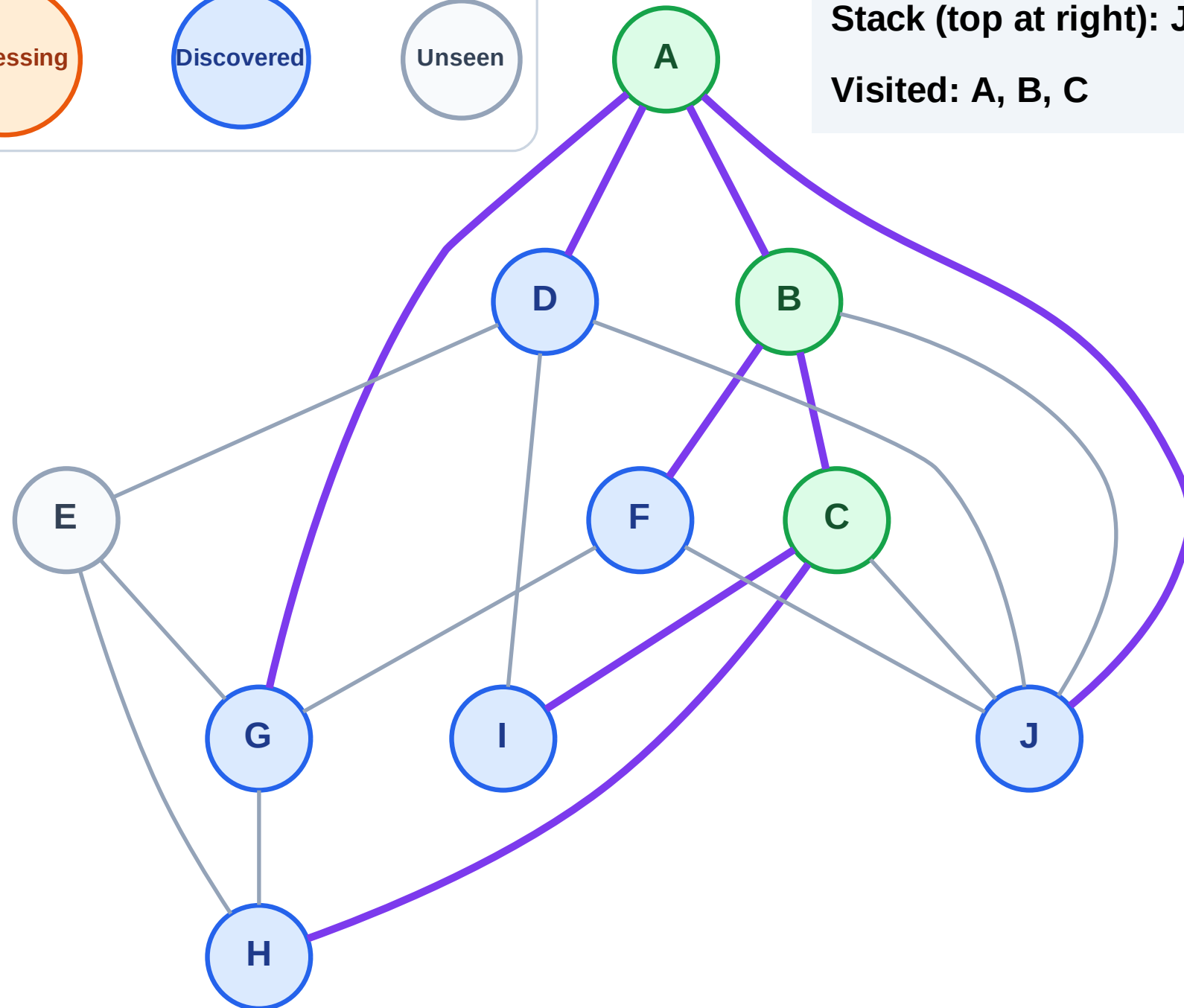
Step 7: Discover I, H from C

## Legend



Stack (top at right): J | G | D | F | I | H

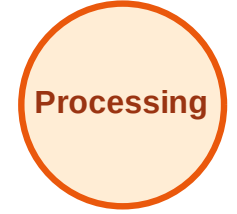
Visited: A, B, C



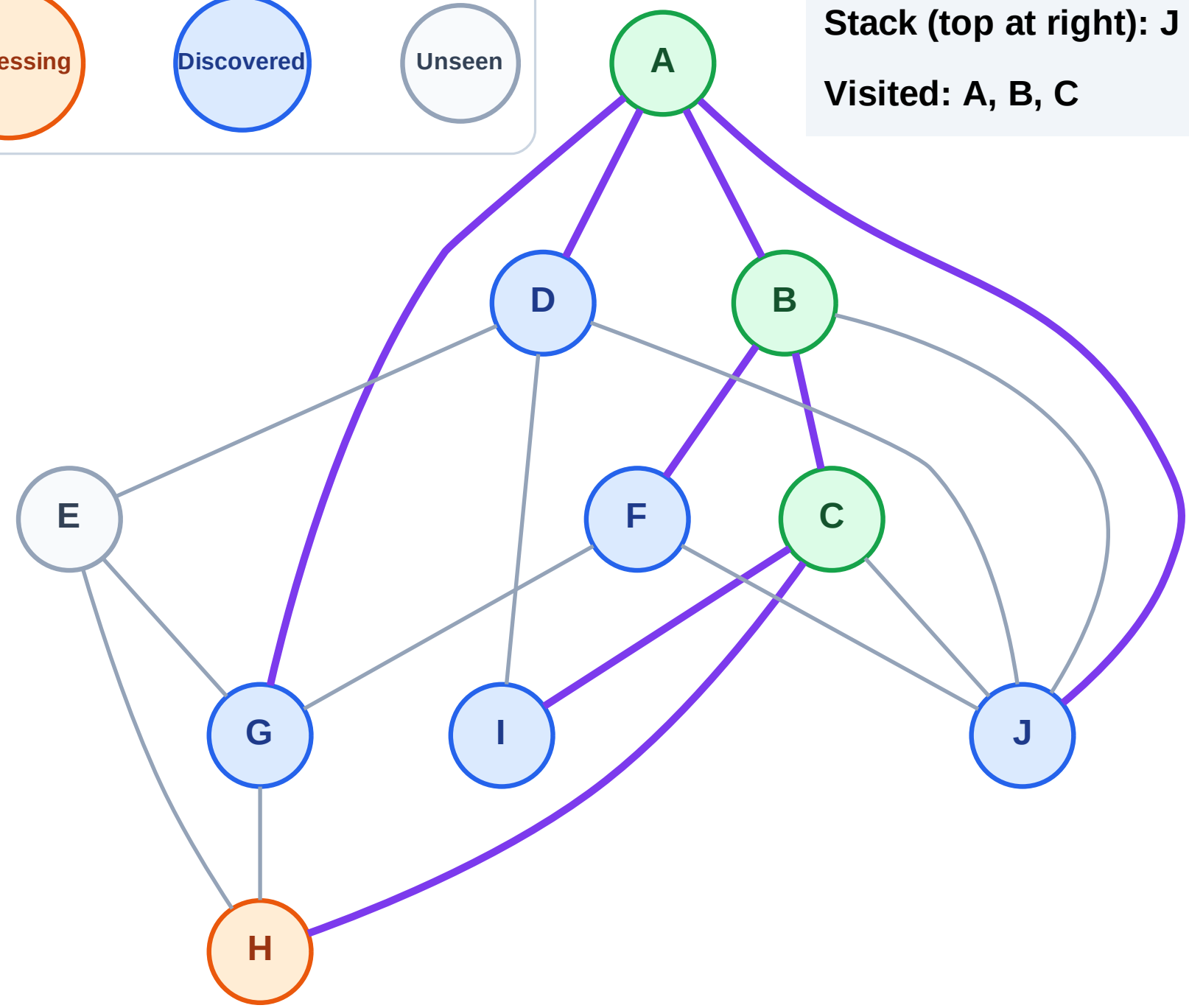
# DFS Traversal

Step 8: Process node H

## Legend



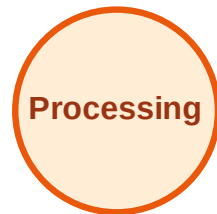
Stack (top at right): J | G | D | F | I  
Visited: A, B, C



# DFS Traversal

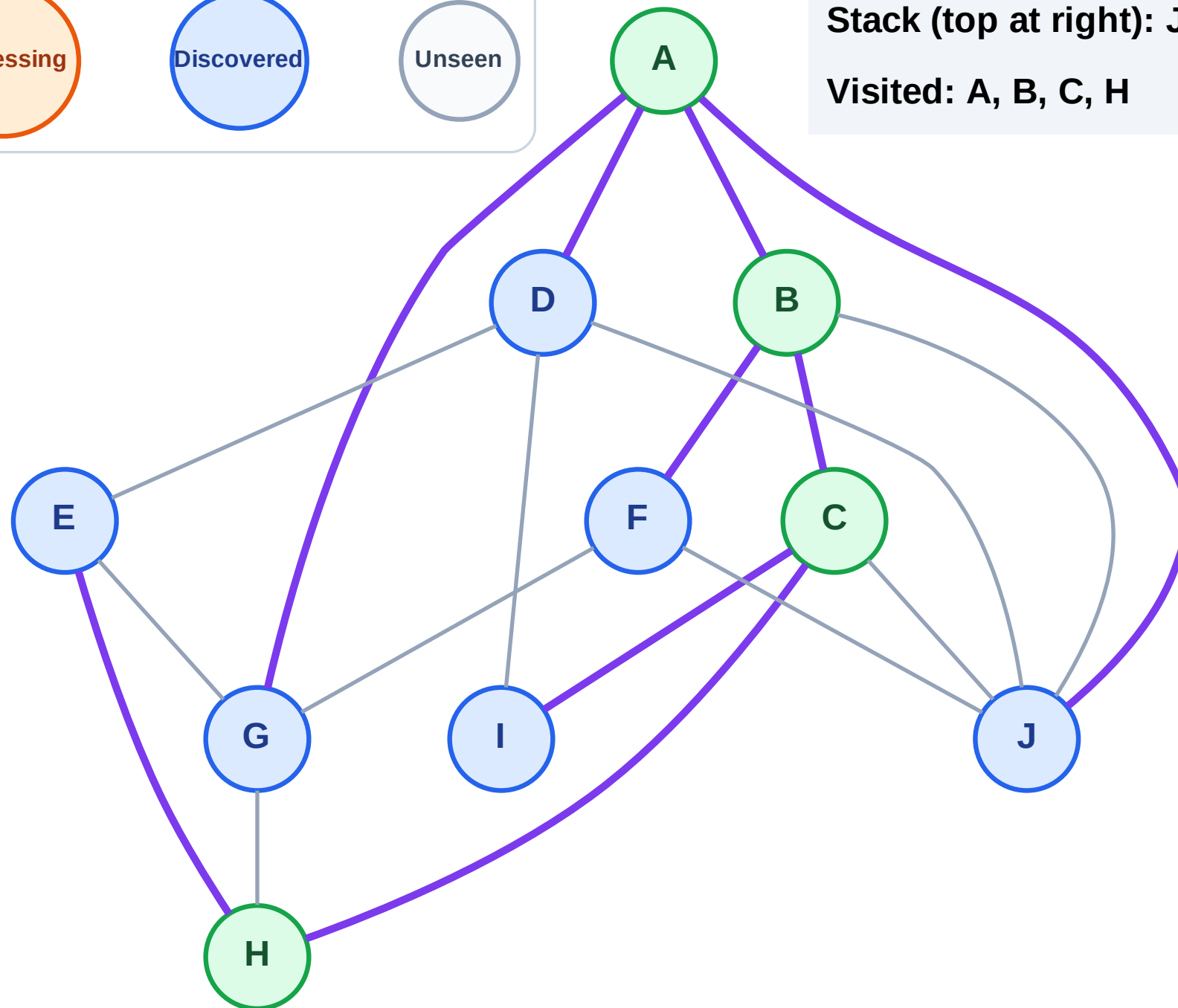
Step 9: Discover E from H

## Legend



Stack (top at right): J | G | D | F | I | E

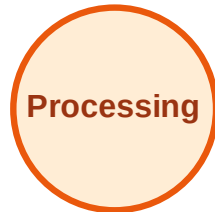
Visited: A, B, C, H



# DFS Traversal

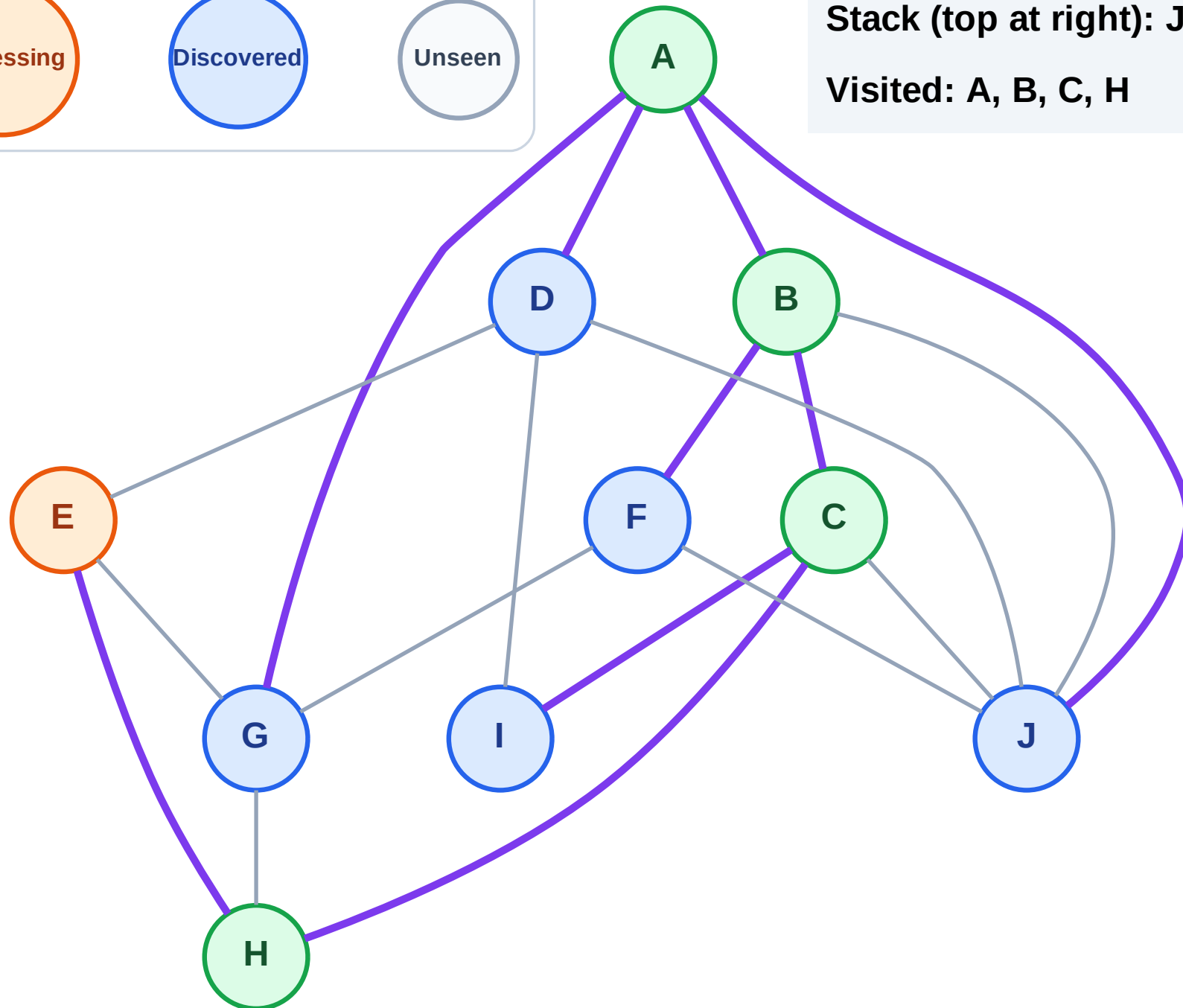
Step 10: Process node E

## Legend



Stack (top at right): J | G | D | F | I

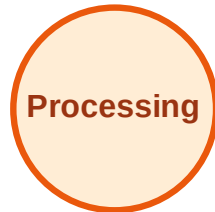
Visited: A, B, C, H



# DFS Traversal

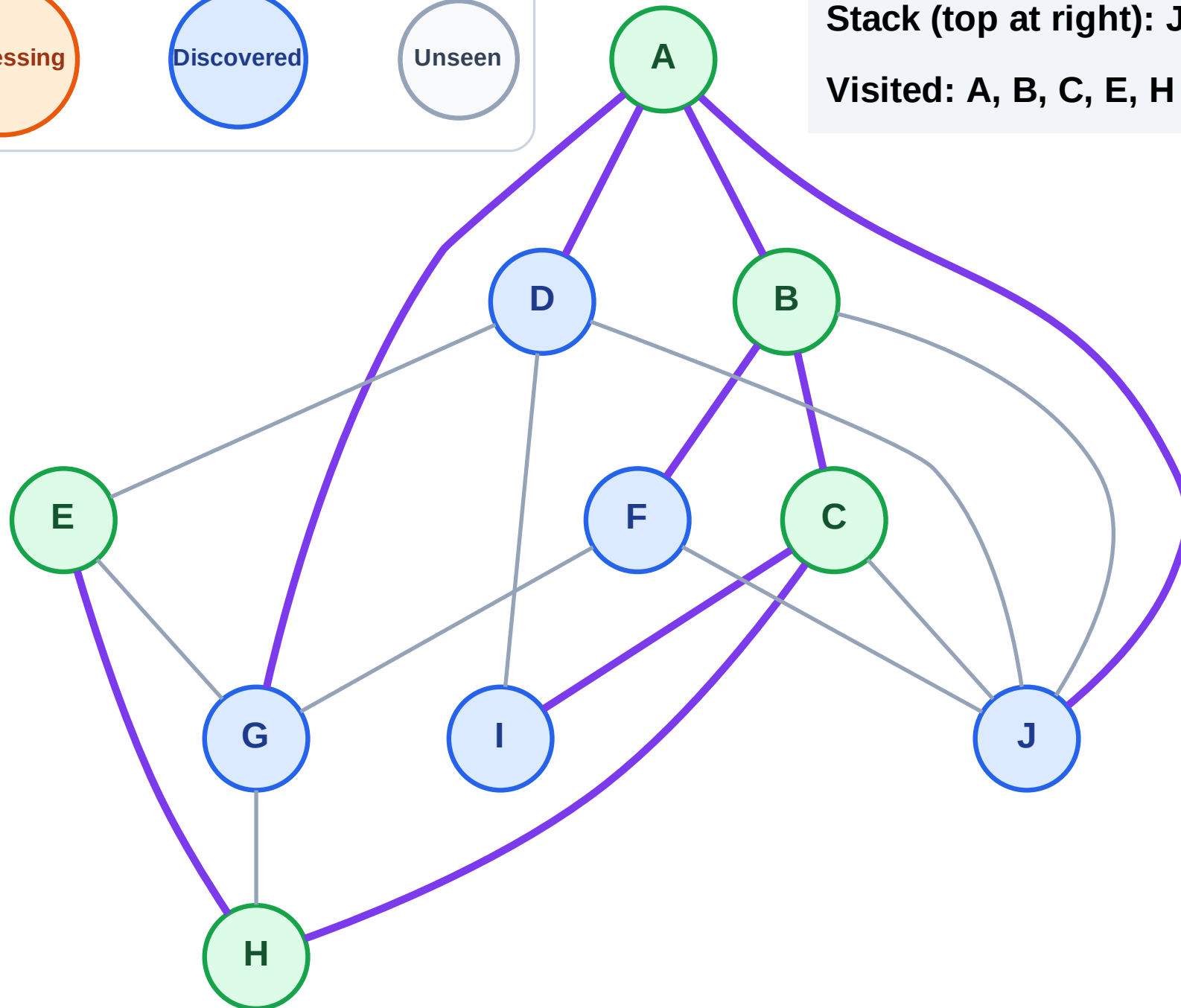
Step 11: No new neighbors from E

## Legend



Stack (top at right): J | G | D | F | I

Visited: A, B, C, E, H





# DFS Traversal

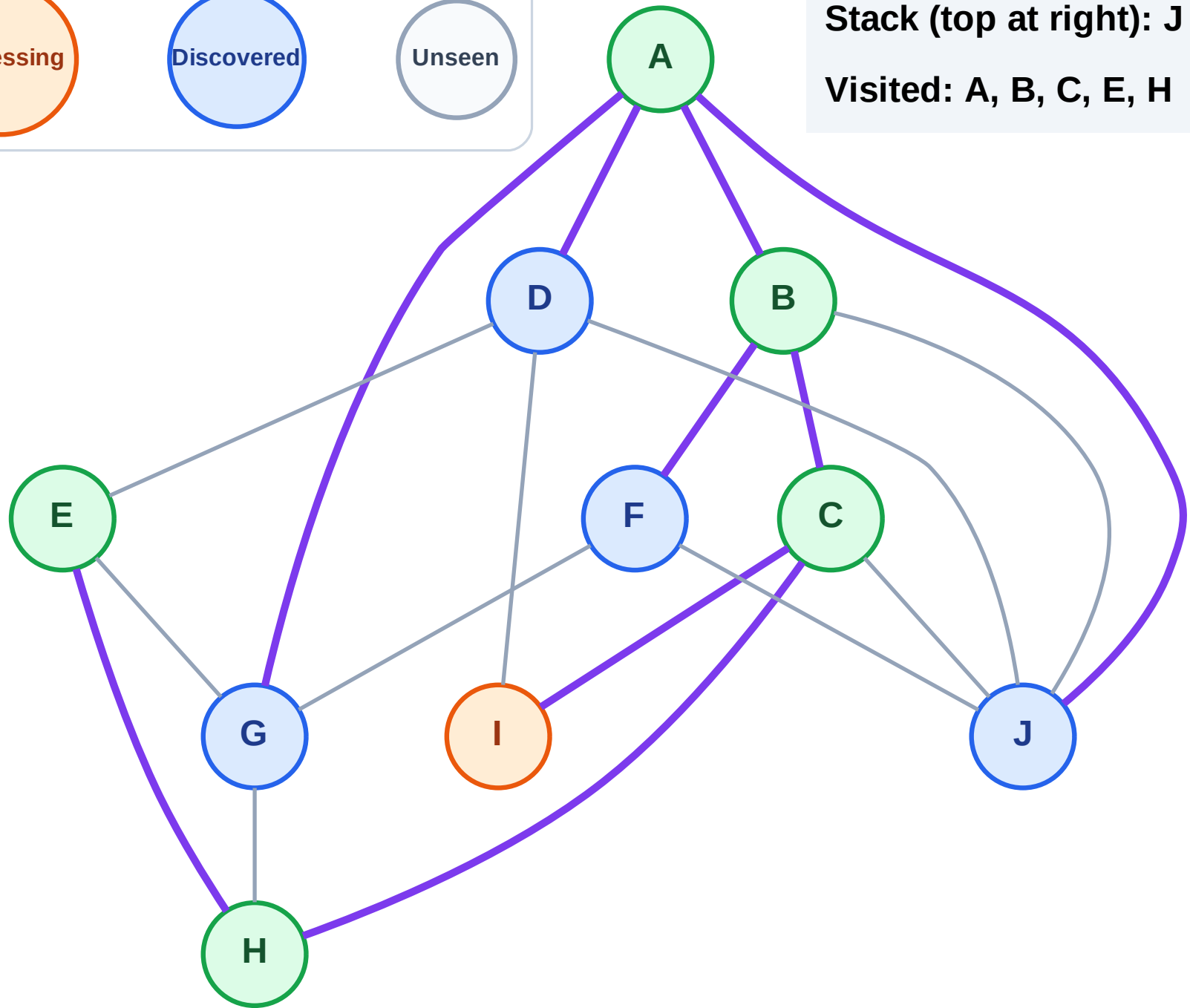
Step 12: Process node I

## Legend



Stack (top at right): J | G | D | F

Visited: A, B, C, E, H



# DFS Traversal

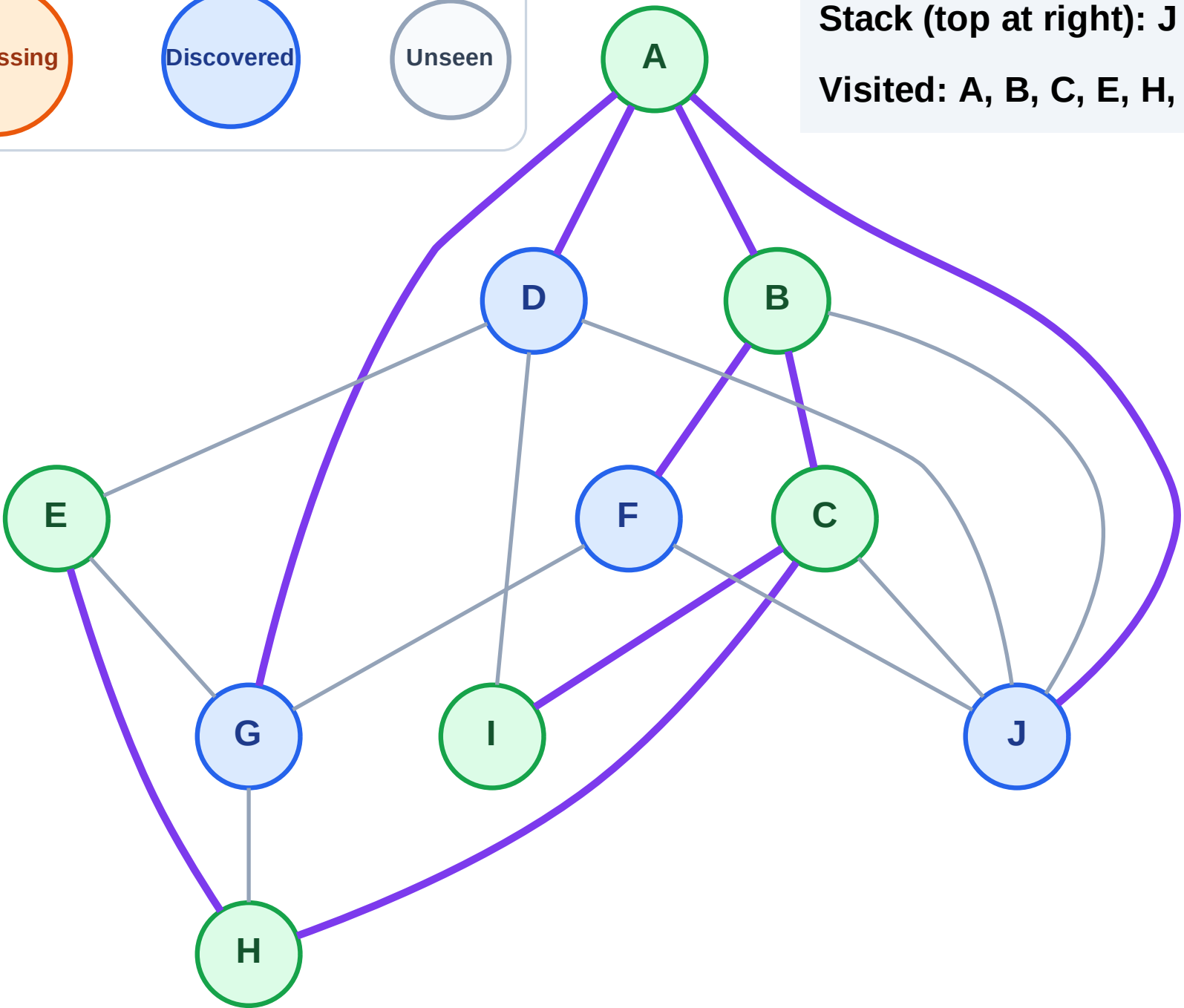
Step 13: No new neighbors from I

## Legend



Stack (top at right): J | G | D | F

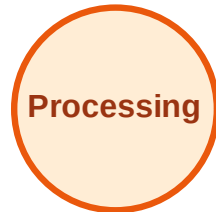
Visited: A, B, C, E, H, I



# DFS Traversal

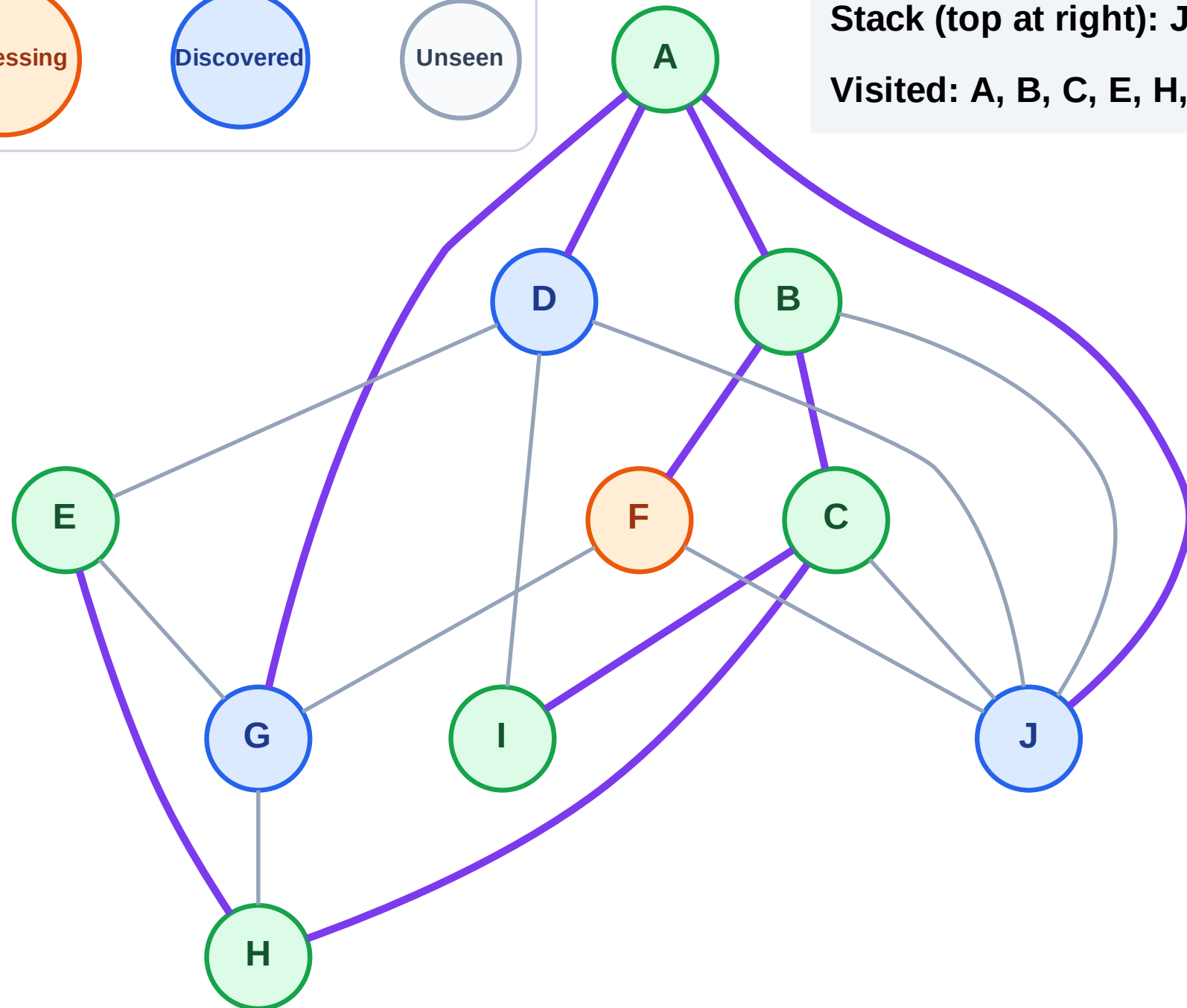
Step 14: Process node F

## Legend



Stack (top at right): J | G | D

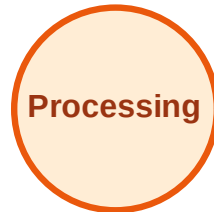
Visited: A, B, C, E, H, I



# DFS Traversal

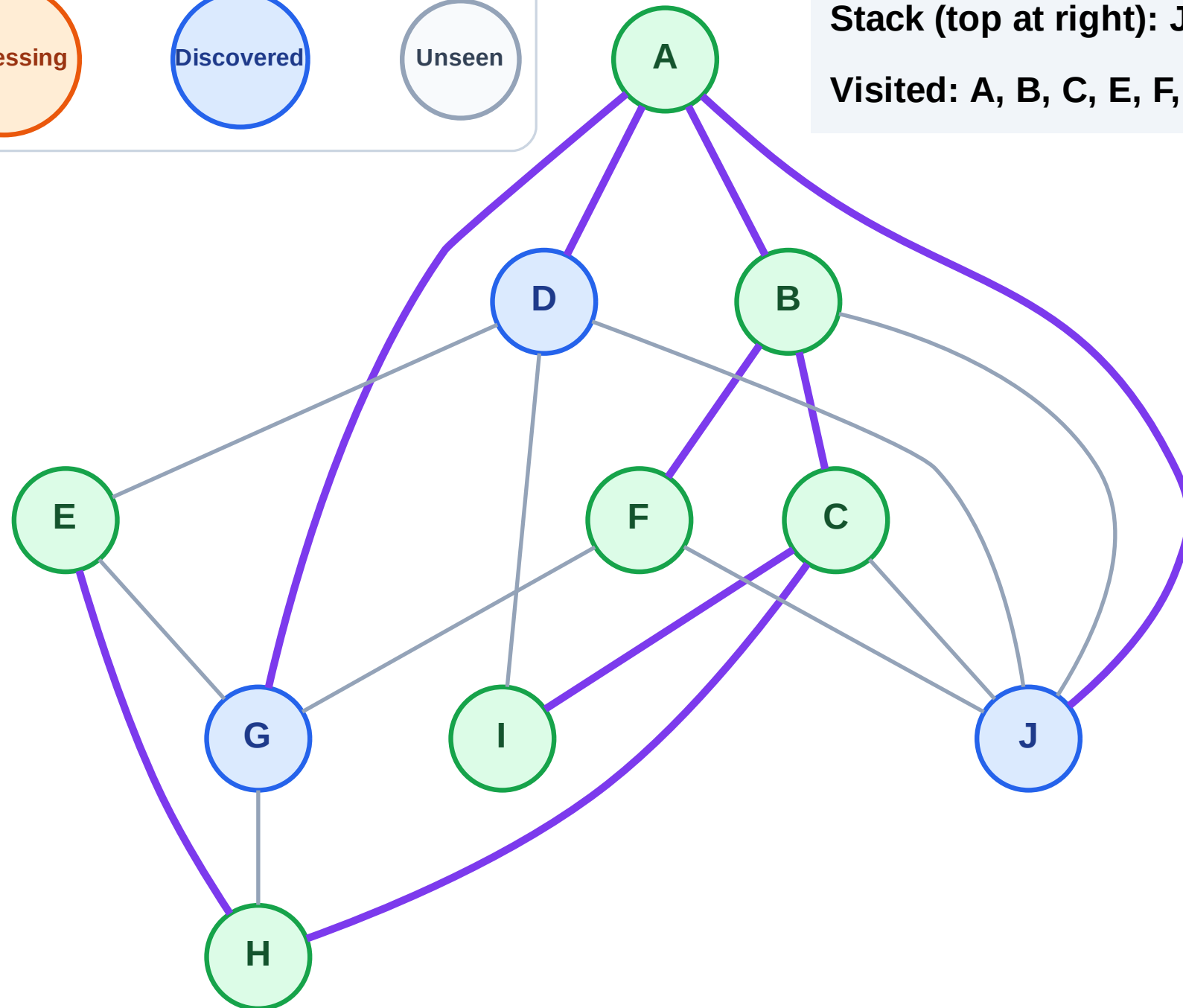
Step 15: No new neighbors from F

## Legend



Stack (top at right): J | G | D

Visited: A, B, C, E, F, H, I



# DFS Traversal

Step 16: Process node D

## Legend

Complete

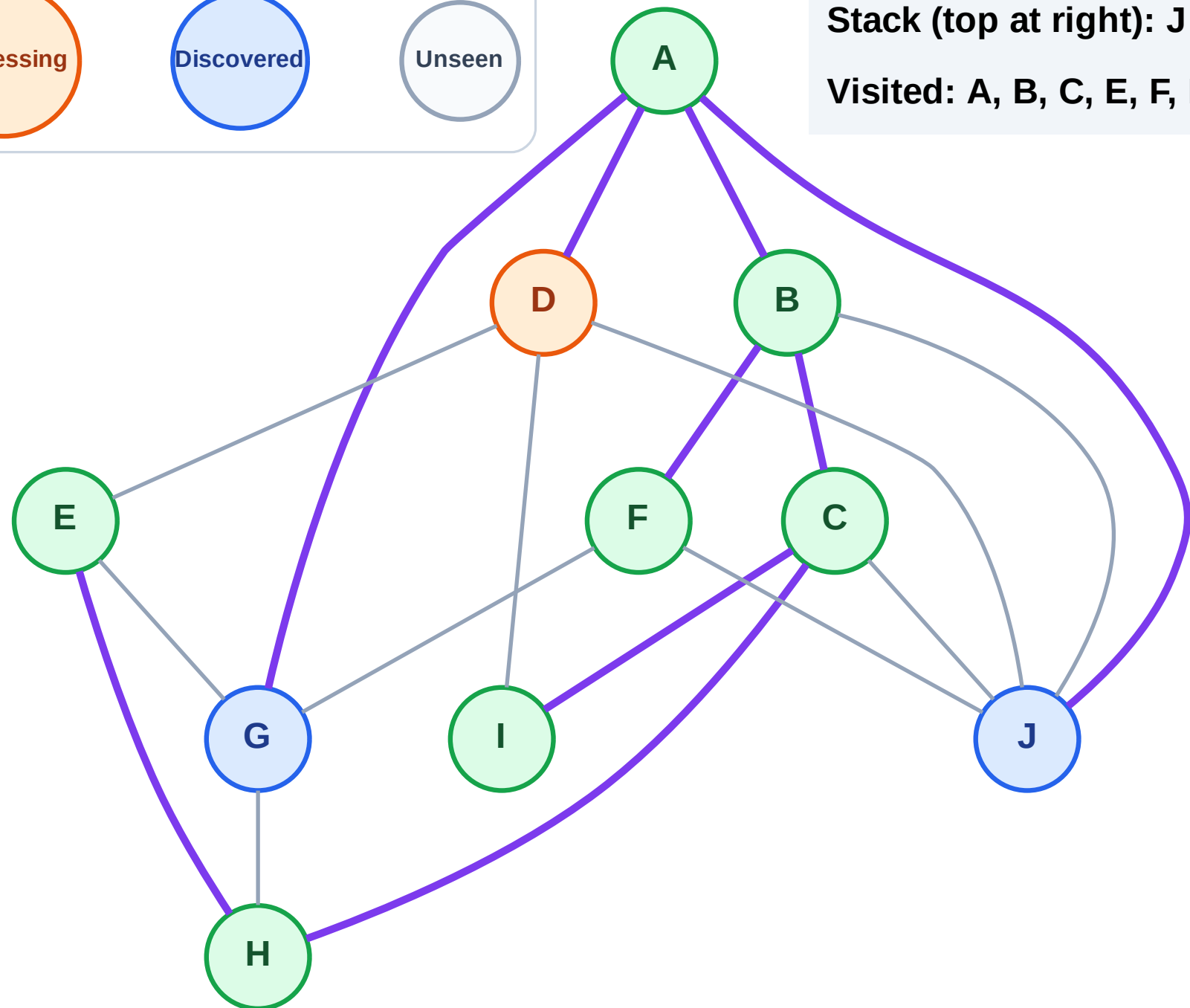
Processing

Discovered

Unseen

Stack (top at right): J | G

Visited: A, B, C, E, F, H, I



# DFS Traversal

Step 17: No new neighbors from D

## Legend

Complete

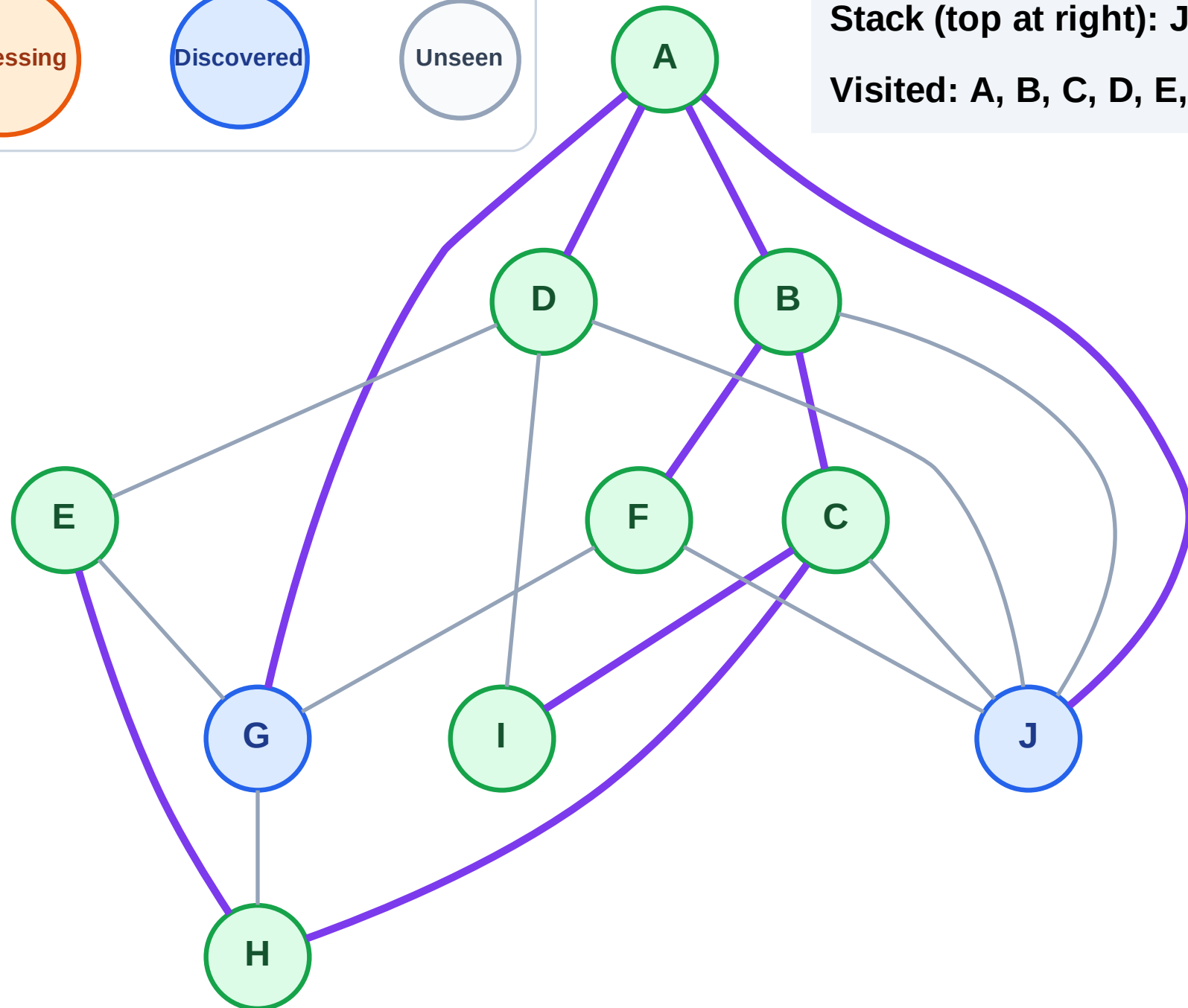
Processing

Discovered

Unseen

Stack (top at right): J | G

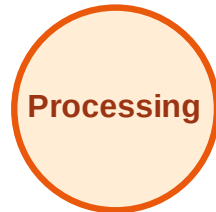
Visited: A, B, C, D, E, F, H, I



# DFS Traversal

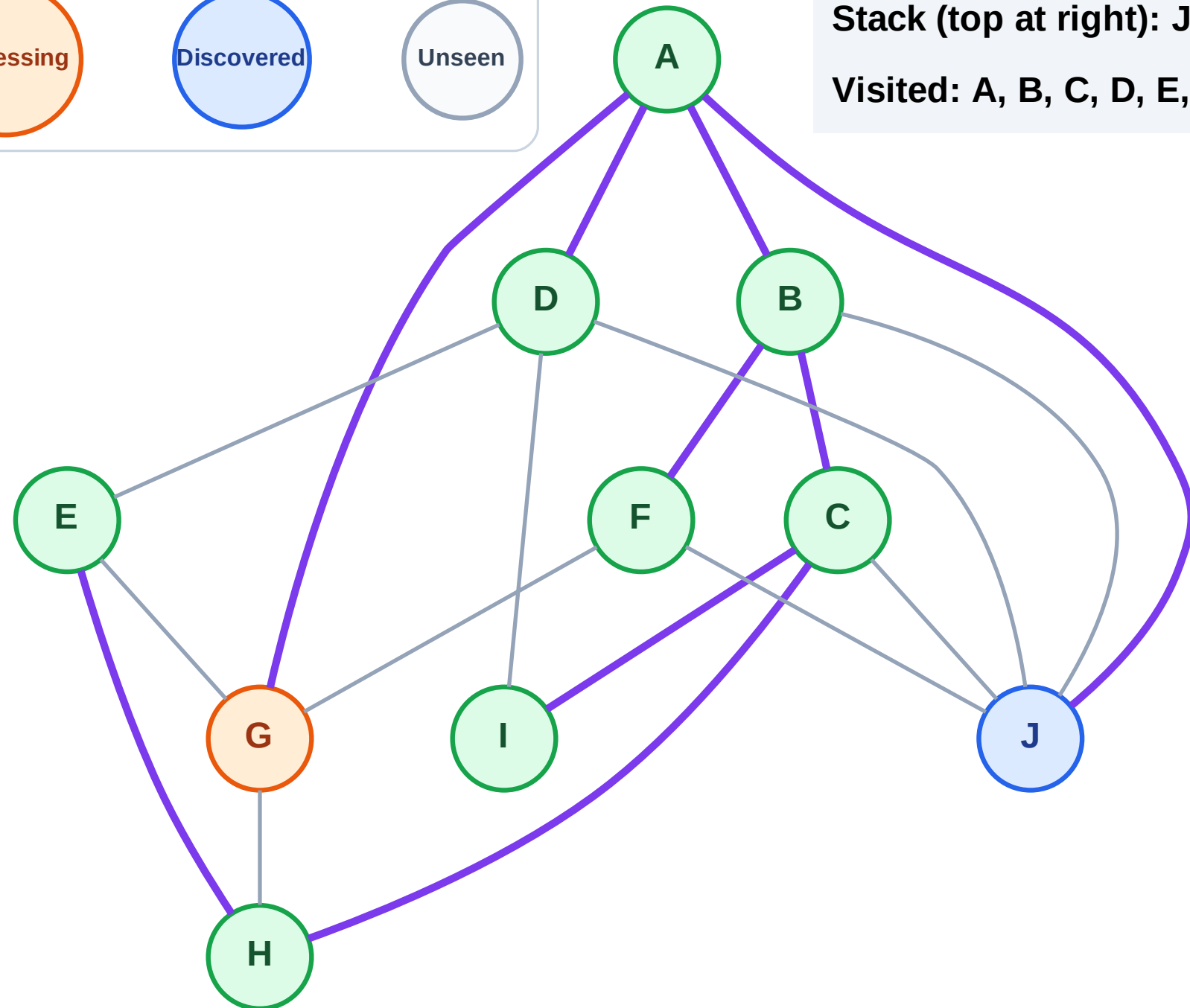
Step 18: Process node G

## Legend



Stack (top at right): J

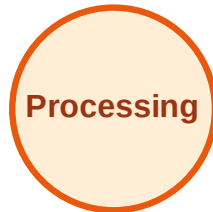
Visited: A, B, C, D, E, F, H, I



# DFS Traversal

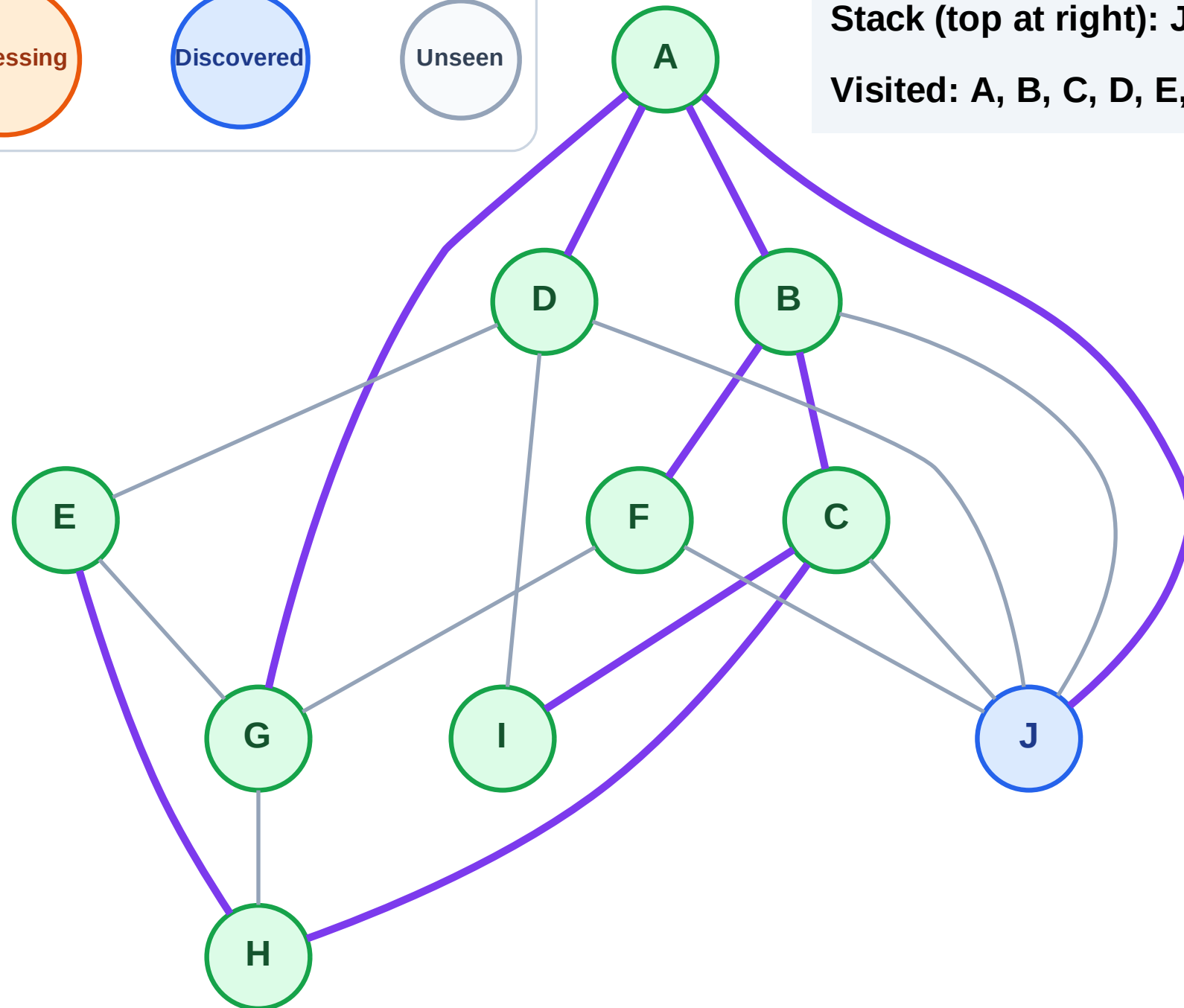
Step 19: No new neighbors from G

## Legend



Stack (top at right): J

Visited: A, B, C, D, E, F, G, H, I

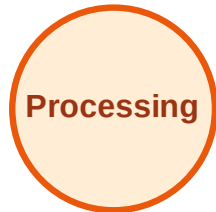




# DFS Traversal

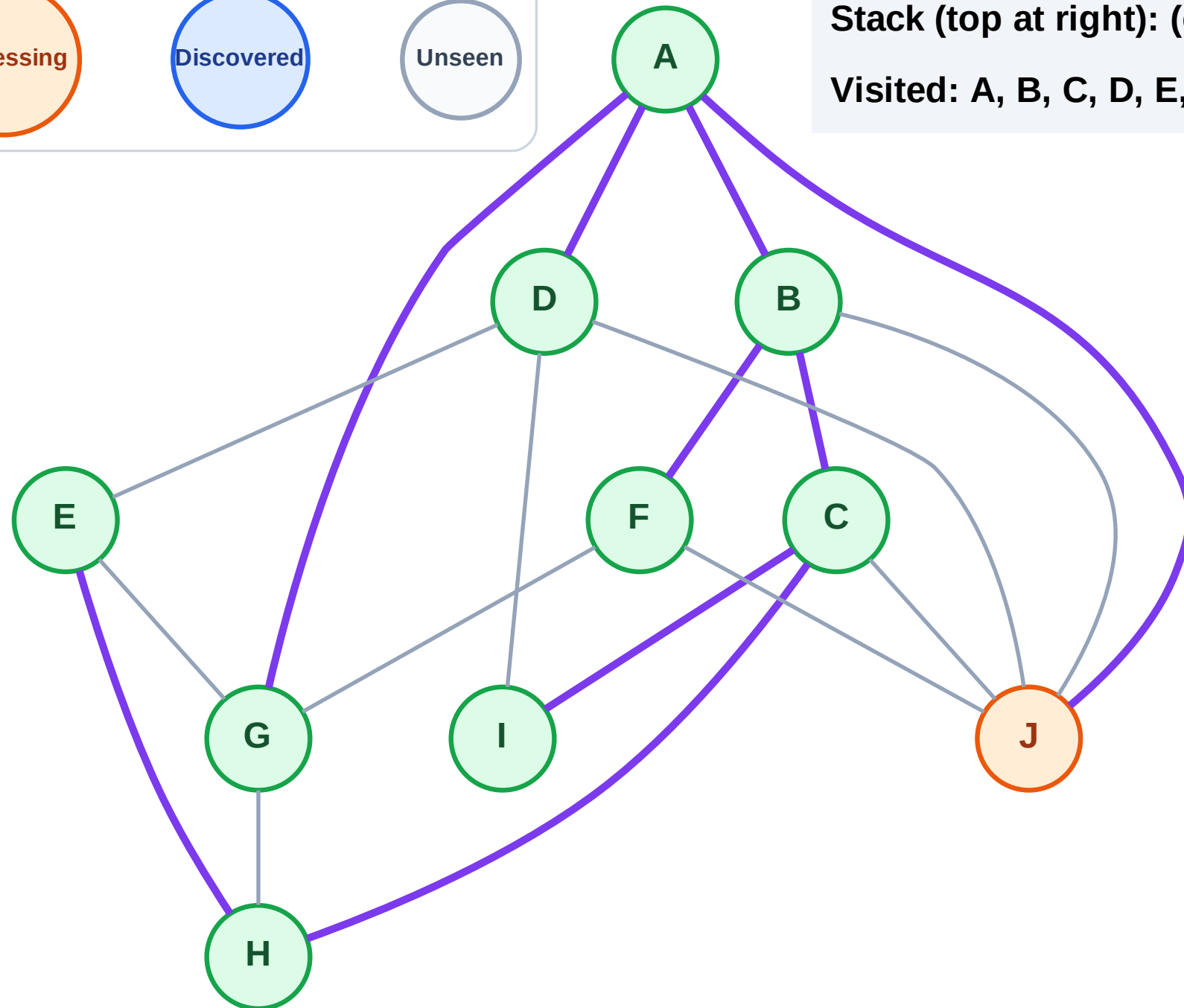
Step 20: Process node J

## Legend



Stack (top at right): (empty)

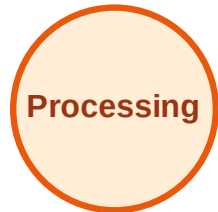
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

